








Modelling Financial Markets

Chapter 6: Multivariate Volatility and Dependence



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When Do Markets Move Together, and How Sure Can We Be?

What we will explain

- Part I: why dependence matters, covariance uncertainty, multivariate GARCH (generalised autoregressive conditional heteroskedasticity) and DCC (dynamic conditional correlation), crisis correlations and the Forbes–Rigobon bias
- Part II: copulas and tail dependence, European and Romanian markets, banks, common volatility and research with artificial intelligence (AI)
- Builds on Chapter 1 (fat tails, volatility clustering), Chapter 4 (Σ in portfolio choice) and Chapter 5 (univariate GARCH)
- Leads to Chapter 7 (portfolio Value at Risk, VaR, and Expected Shortfall, ES), Chapter 8 (backtesting) and Chapter 18 (systemic risk)

What you should be able to do

- estimate a time-varying covariance (rolling window, EWMA: exponentially weighted moving average, DCC) and attach a valid standard error to a correlation
- separate contagion from the volatility bias of crisis correlations
- choose, fit and test a copula, and read its tail dependence at a finite threshold and in the limit

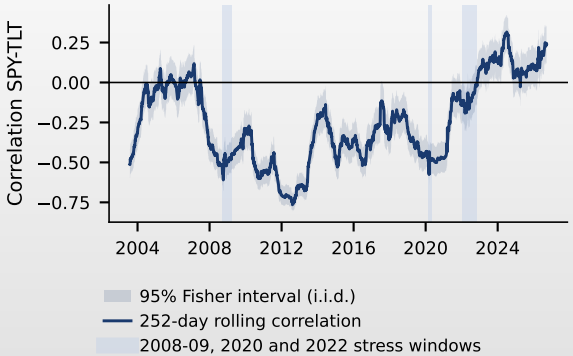
Why Dependence Matters (1/2): Portfolio Diversification

- **Diversification** (Markowitz (1952))
  - The portfolio variance depends on the correlation through a single cross term
  - $\sigma_p^2 = w^\top \Sigma w = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\,\sigma_1\sigma_2$
  - w: weight of asset 1 ($1-w$ in asset 2)
  - Σ: covariance matrix
  - σ_1, σ_2: volatilities
  - ρ: correlation
  - Portfolio 60/40 of SPY (S&P 500 exchange-traded fund) and TLT (20+ year US Treasury bond fund)
  - Same asset volatilities, $\sigma_1 = 18.8\%$ (SPY) and $\sigma_2 = 14.3\%$ (TLT): daily standard deviations over 2002–2026 times $\sqrt{252}$, weights reset to 60/40 every day
  - Portfolio volatility 10.4% with the 2002–2021 correlation (-0.40), 13.2% with the 2022–2026 one (0.11). Check: $0.6^2 \cdot 0.188^2 + 0.4^2 \cdot 0.143^2 + 2 \cdot 0.6 \cdot 0.4 \cdot (-0.40) \cdot 0.188 \cdot 0.143 = 0.0108$, whose square root is 10.4%

Why Dependence Matters (2/2): Hedges, Contagion and Moving Covariances

- **Hedging**
  - How much value of the hedge F to sell per unit of value held in the position S to minimise the variance
  - Minimum-variance hedge ratio $h^* = \rho\,\sigma_S/\sigma_F$: the slope of the regression of r_S on r_F. Estimated on returns, it is a ratio of values: units of F per unit of S are $h^* P_S/P_F$ (P_S, P_F: prices, contract size included)
  - r_S, r_F: returns of the position and of the hedge
  - σ_S, σ_F: their volatilities
  - Effectiveness ρ^2: the share of the variance removed
  - If the correlation moves, yesterday's hedge is wrong today: a dynamic hedge needs a dynamic ρ_t
- **Contagion and interdependence** (Forbes & Rigobon (2002))
  - Contagion: a *significant increase* in cross-market linkages after a shock
  - Interdependence: links present in all states (trade, common investors, common macro factors)
  - Correlations tend to rise exactly when diversification is needed most
- New in this chapter: Σ_t, the covariance matrix at date t, moves
- How to estimate it, and how to measure the sampling uncertainty of every dependence estimate

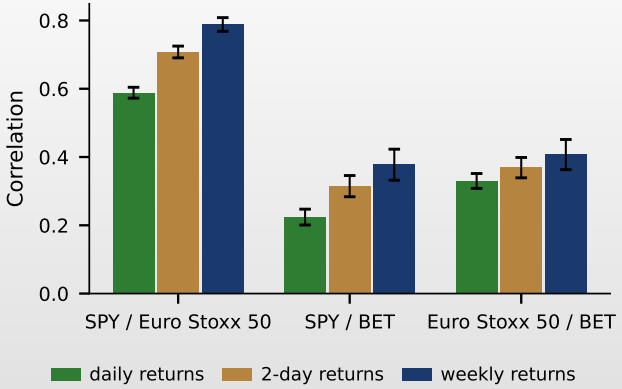
Stocks and Bonds: A Correlation That Changed Sign



- Daily SPY and TLT returns, 252-day window
- Band: 95% Fisher interval assuming i.i.d. (independent and identically distributed) data
- Shaded: autumn 2008 – March 2009, February–April 2020, January–October 2022
- Minimum −0.77 (8 August 2012), maximum 0.32 (1 July 2024), last 0.24 (18 September 2026)
- Positive on 6% of days before 2022, on 81% since January 2022
- Growth shocks move stocks and bond prices in opposite directions (negative correlation)
- Inflation shocks hit both (positive, as in 2022)
- The Fisher band is too narrow here: it ignores fat tails and volatility clustering (next section)

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Daily, 2-Day and Weekly Correlations, 2000–2026 (1/2)



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Daily, 2-Day and Weekly Correlations, 2000–2026 (2/2)

- SPY, Euro Stoxx 50 and BET (the main index of the Bucharest Stock Exchange) on common trading days
- Vertical lines on the bars: 95% Fisher intervals
- SPY–Euro Stoxx 50: 0.59 daily, 0.71 on 2-day returns, 0.79 weekly
- SPY yesterday with Euro Stoxx 50 today: 0.20
- SPY–BET: 0.22 daily versus 0.38 weekly (lagged 0.17)
- Euro Stoxx 50–BET, almost synchronous closes: 0.33 versus 0.41
- Part of the "low" US–Europe daily correlation is a measurement artefact
- Weekly data are cleaner but give 1 367 instead of 6 446 observations: wider intervals

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Asynchronous Trading: A Trap for Daily Correlations

- Closing times differ: Bucharest and Frankfurt close hours before New York
  - US news after the European close enters European prices only the next day
- Consequence (positive dependence, as between equity markets): the same-day correlation is biased towards zero, and the lagged correlation is positive
- Remedies
  - Lower frequency: 2-day or weekly returns on common trading days
  - Add the lead–lag term: $\rho_0 + \rho_{+1}$ (ρ_0: same-day correlation, ρ_{+1}: US yesterday with Europe today)
  - A rough approximation of the synchronous correlation, valid only for a one-day lag, no autocorrelation and equal variances
- Rule used throughout: first align the *prices* on common trading days, then compute returns

Three Frequencies, Each Chosen for a Reason

- Returns are log returns, $r_t = \ln P_t - \ln P_{t-1}$, computed after aligning the prices on common trading days
- Annualised volatilities multiply the daily standard deviation by $\sqrt{252}$
- Daily: DCC, HAC intervals and hedging, where the reaction speed matters
- 2-day: the Forbes–Rigobon tests, to absorb different closing times
- Consecutive 2-day returns overlap by one day
- Weekly (Friday closes): exceedance correlations, market integration and bank networks, which mix time zones

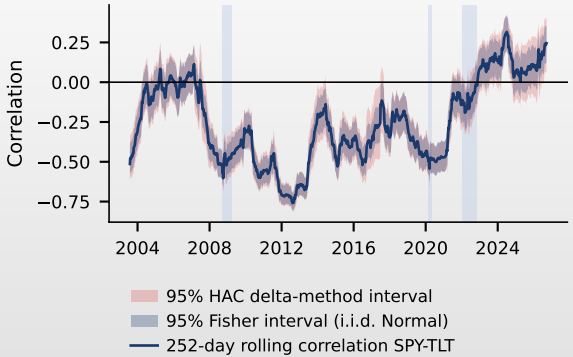
Sample Covariance and Inference on a Correlation (1/2): Estimator and Fisher Interval

- Sample covariance matrix and Pearson correlation of N assets over T days
  - $\hat{\Sigma} = \frac{1}{T-1} \sum_t (r_t - \bar{r})(r_t - \bar{r})^\top, \quad \hat{\rho}_{ij} = \hat{\Sigma}_{ij} / \sqrt{\hat{\Sigma}_{ii} \hat{\Sigma}_{jj}}$
  - $r_t \in \mathbb{R}^N$: vector of the N returns on day t, $t = 1, \ldots, T$
  - $\bar{r}$: vector of sample means
  - $\hat{\Sigma}_{ij}$: element (i, j)
  - $N(N+1)/2$ distinct elements to estimate: 125 250 for $N = 500$
- **Fisher transform**: $z = \operatorname{artanh} \hat{\rho} = \frac{1}{2} \ln \frac{1+\hat{\rho}}{1-\hat{\rho}}$ is approximately Normal with a variance that does not depend on ρ
  - $z \approx N\big(\operatorname{artanh} \rho,\ 1/(T-3)\big)$ for i.i.d. pairs with the bivariate Normal distribution
  - 95% interval $\tanh(z \pm 1.96/\sqrt{T-3})$, mapped back to the correlation scale by tanh
  - $T = 252$, $\rho = 0.5$: half-width about 0.09

Sample Covariance and Inference on a Correlation (2/2): Delta Method and HAC

- The delta method writes the estimation error $\hat{\rho} - \rho$ as a sample mean of daily contributions ψ_t
- Volatility clustering does not change this form, only the variance of ψ_t
  - $\hat{\rho} - \rho \approx \frac{1}{T} \sum_t \psi_t, \quad \psi_t = \tilde{x}_t \tilde{y}_t - \frac{\rho}{2}(\tilde{x}_t^2 + \tilde{y}_t^2)$
  - $\tilde{x}_t$, $\tilde{y}_t$: the two returns standardised to mean 0 and variance 1
  - ψ_t: the contribution of day t to the error
- Standard error (se) from the long-run variance of ψ_t
  - HAC (heteroskedasticity and autocorrelation consistent): Bartlett kernel, L lags (Newey & West (1987))
  - Needs stationarity, finite fourth moments and weak dependence
  - Lags by the usual rule of thumb $L = \lfloor 4(T/100)^{2/9} \rfloor$: $L = 9$ for $T = 4\,891$, $L = 6$ for $T = 1\,182$, $L = 4$ for a 252-day window
  - i.i.d. Normal pairs give $\mathrm{Var}(\psi_t) = (1 - \rho^2)^2$, the Fisher variance
  - Fat tails and GARCH raise it (derivation in the Appendix)
- SPY–TLT daily, 2002–2021 ($T = 4\,891$, $\hat{\rho} = -0.403$): se 0.0120 (Fisher), 0.0217 (fourth moments, $L = 0$), 0.0227 (HAC, $L = 9$)

Fisher versus HAC Intervals: SPY and TLT



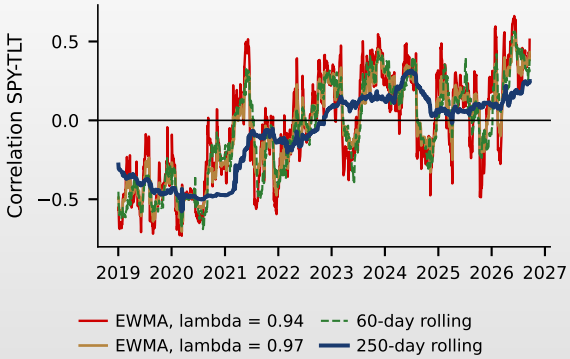
- 252-day rolling correlation, every 5th day, with the 95% Fisher interval and the 95% HAC delta-method interval (Bartlett, $L = 4$)
- HAC is wider in almost every window: median width ratio 1.24, 90% quantile 1.54
- Correlation significantly different from zero: 73% of windows by Fisher, 65% by HAC
- The verdict flips in 8.2%
- Whole periods: HAC se $= 1.9\times$ Fisher in 2002–2021, $1.4\times$ since 2022
- Fourth moments alone give $1.8\times$: fat tails matter more than autocorrelation
- Rule for daily returns: report delta-method HAC intervals

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Rolling Windows and EWMA

- **Rolling window** of length m: $\hat{\Sigma}_t = \frac{1}{m}\sum_{s=t-m}^{t-1} r_s r_s^\top$ (mean set to zero for daily data)
  - r_s: return vector of day s
  - m: number of past days used
  - Positive semidefinite (every portfolio variance ≥ 0): for any weight vector v, $v^\top \hat{\Sigma}_t v = \frac{1}{m}\sum_s (v^\top r_s)^2 \geq 0$
  - Singular if $m < N$
  - Positive definite only if the m return vectors span $\mathbb{R}^N$
  - **Ghost effect**: an extreme day enters with weight $1/m$ and leaves abruptly m days later
- **EWMA** (exponentially weighted moving average, J.P. Morgan/Reuters (1996)):
  $\Sigma_t = \lambda \Sigma_{t-1} + (1-\lambda) r_{t-1} r_{t-1}^\top = (1-\lambda)\sum_{k\geq 1} \lambda^{k-1} r_{t-k} r_{t-k}^\top$
  - $\lambda \in (0,1)$: decay factor
  - k: age of the observation in periods, weight $(1-\lambda)\lambda^{k-1}$
  - $\lambda = 0.94$ for daily and 0.97 for monthly data
  - Half-life $\ln 0.5 / \ln \lambda$: 11.2 and 22.8 periods
  - Effective sample $(1+\lambda)/(1-\lambda) \approx 32$ days for $\lambda = 0.94$
  - Positive semidefinite, no ghost effect
  - But one λ for all elements and no mean reversion: an integrated GARCH(1,1) without a constant (Chapter 5)
- One trade-off for both: a short window or a small λ reacts fast but is noisy
- A long window or a large λ is stable but slow to detect a regime change

EWMA versus Rolling Windows, SPY–TLT since 2019



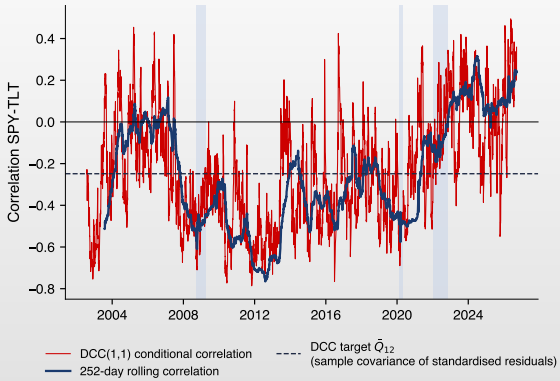
- Two EWMA estimators ($\lambda = 0.94$ and 0.97) and two rolling windows (60 and 250 days) on the same data
- Standard deviation of the daily change of the estimate: 0.060 (EWMA 0.94), 0.032 (EWMA 0.97), 0.0088 (250-day window)
- From 10 May 2022 the EWMA correlation with $\lambda = 0.94$ stays positive for 20 consecutive days
- The 250-day window gives the same signal only on 8 December 2022
- With EWMA and rolling windows, λ and m are chosen by the analyst, not estimated
- DCC (next section) estimates the reaction speed and the memory of the correlation from the data

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From Univariate to Multivariate Volatility (1/2): the Model

- Chapter 5: one asset, $r_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$
- Now N assets and a whole covariance matrix
  - $r_t = H_t^{1/2} z_t, \quad \mathbb{E}(z_t \mid \mathcal{F}_{t-1}) = 0, \quad \text{Var}(z_t \mid \mathcal{F}_{t-1}) = I_N, \quad \text{so Var}(r_t \mid \mathcal{F}_{t-1}) = H_t$
  - r_t, z_t: vectors of N returns and innovations
  - H_t: $N \times N$ conditional covariance matrix, known at $t-1$ (Σ_t in the previous section)
  - $H_t^{1/2}$: a matrix square root, $H_t^{1/2}(H_t^{1/2})^\top = H_t$
  - I_N: identity matrix
  - $\mathcal{F}_{t-1}$: information up to $t-1$
- Any model for H_t must deliver
  - positive definiteness at every t (every portfolio has a positive variance)
  - Few parameters
  - Interpretable dynamics
- Gaussian quasi-log-likelihood, maximised over the parameters of H_t
  - $\ell = -\frac{1}{2} \sum_t \left(N \ln 2\pi + \ln |H_t| + r_t^\top H_t^{-1} r_t \right)$
  - $|H_t|$: determinant of H_t
  - Quasi: the estimates stay consistent if z_t is not Normal but H_t is right

DCC Correlation of SPY and TLT, 2002–2026



- DCC(1,1) conditional correlation, the 252-day rolling correlation and the target $\bar{Q}_{12}$, the long-run correlation of the model (next slides)
- Range −0.79 (2 November 2011) to 0.49 (21 May 2026)
- Last value 0.36
- Average: −0.54 from 15 September to 31 December 2008, −0.56 in 2012, 0.00 in 2022
- DCC reacts within days, like EWMA, but its speed a and memory b are estimated
- Since January 2022 the path is above the target on 92% of days (36% before): a single long-run correlation is doubtful

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From Univariate to Multivariate Volatility (2/2): the Curse of Dimension

- **VEC** (vectorised GARCH, Bollerslev, Engle & Wooldridge (1988))
  - Every element of H_t on all squares and cross-products
  - Positive definiteness not guaranteed
- **BEKK** (Baba–Engle–Kraft–Kroner, Engle & Kroner (1995))
  - Positive definite by construction, hard to interpret (recursions in the Appendix)
- **DCC**: $3N$ univariate GARCH parameters plus 2
  - The correlation target is estimated separately (next slide)
- The number of parameters of VEC and BEKK grows like N^4 and N^2: infeasible for large portfolios

Number of parameters

N	VEC	full BEKK	DCC
2	21	11	$6+2$
5	465	65	$15+2$
10	6 105	255	$30+2$
50	3 252 525	6 275	$150+2$

Constant and Dynamic Conditional Correlation

- Decomposition: covariance = volatilities $\times$ correlations $\times$ volatilities
  - $H_t = D_t R_t D_t, \quad D_t = \mathrm{diag}(\sigma_{\mathbf{1}t}, \ldots, \sigma_{Nt})$
  - σ_{it}: GARCH volatility of asset i
  - R_t: conditional correlation matrix
  - $\varepsilon_{it} = r_{it}/\sigma_{it}$: standardised residuals
- **CCC** (constant conditional correlation, Bollerslev (1990)): $R_t = R$ constant
- **DCC** (Engle (2002)): a GARCH-type recursion for a proxy matrix Q_t
  - $Q_t = (1 - a - b)\,\bar{Q} + a\,\varepsilon_{t-\mathbf{1}}\varepsilon_{t-\mathbf{1}}^{\top} + b\,Q_{t-\mathbf{1}}, \quad a, b \geq 0,\ a + b < 1$
  - $R_t = \mathrm{diag}(Q_t)^{-\mathbf{1/2}}\, Q_t\, \mathrm{diag}(Q_t)^{-\mathbf{1/2}}$
  - $\bar{Q}$: the intercept matrix of the recursion, with $N(N-1)/2$ free elements. Correlation targeting replaces it by the sample covariance $\hat{\bar{Q}} = \frac{\mathbf{1}}{T}\sum_t \varepsilon_t\varepsilon_t^{\top}$ (when this is consistent: slide "Two-Step Estimation (2/2)")
  - a: reaction to yesterday's product of shocks
  - b: memory
  - $a + b$: persistence
  - The second line rescales Q_t so that its diagonal is 1: R_t is a valid correlation matrix
- Two parameters govern all $N(N-1)/2$ correlations: parsimonious, but every pair has the same persistence
- Q_t is a covariance proxy, not a correlation: $\mathbb{E}_{t-\mathbf{1}}[\varepsilon_t\varepsilon_t^{\top}] = R_t \neq Q_t$

Two-Step Estimation (1/2): Splitting the Likelihood

- The Gaussian log-likelihood splits into a volatility part and a correlation part
  - $\ell = \underbrace{-\frac{1}{2}\sum_t \left(N \ln 2\pi + 2\ln|D_t| + r_t^\top D_t^{-2} r_t\right)}_{\ell_V} \underbrace{-\frac{1}{2}\sum_t \left(\ln|R_t| + \varepsilon_t^\top R_t^{-1}\varepsilon_t - \varepsilon_t^\top \varepsilon_t\right)}_{\ell_C}$
  - ℓ_V: depends only on the GARCH parameters (volatility part)
  - ℓ_C: depends on (a, b) given the volatilities (correlation part)
  - $|D_t|$, $|R_t|$: determinants
- Where the split comes from: substitute $H_t = D_t R_t D_t$ and $\varepsilon_t = D_t^{-1} r_t$ into the quasi-log-likelihood of the previous section
  - Determinant: $|H_t| = |D_t|^2 |R_t|$, so $\ln|H_t| = 2\ln|D_t| + \ln|R_t|$
  - Quadratic form: $r_t^\top H_t^{-1} r_t = r_t^\top D_t^{-1} R_t^{-1} D_t^{-1} r_t = \varepsilon_t^\top R_t^{-1} \varepsilon_t$
  - Add and subtract $\varepsilon_t^\top \varepsilon_t = r_t^\top D_t^{-2} r_t = \sum_i r_{it}^2/\sigma_{it}^2$: then $\ell_V = \sum_i \left[-\frac{1}{2}\sum_t (\ln 2\pi + \ln\sigma_{it}^2 + r_{it}^2/\sigma_{it}^2)\right]$, the sum of the N univariate Gaussian GARCH log-likelihoods
- Step 1: fit N univariate GARCH(1,1) models, keep σ_{it} and ε_{it}
- Step 2: fix $\bar{Q}$ at the sample covariance of ε_t and maximise ℓ_C over (a, b)

Two-Step Estimation (2/2): Targeting and Inference

- Targeting with the sample covariance makes the standard DCC estimator inconsistent in general. Why: take expectations of the recursion, using $\mathbb{E}[\varepsilon_{t-\mathbf{1}}\varepsilon_{t-\mathbf{1}}^\top] = \mathbb{E}[R_{t-\mathbf{1}}]$
  - Stationarity gives $(1-b)\,\mathbb{E}[Q_t] = (1-a-b)\,\bar{Q} + a\,\mathbb{E}[R_t]$
  - $\hat{\bar{Q}} = \frac{\mathbf{1}}{T}\sum_t \varepsilon_t\varepsilon_t^\top$ estimates $\mathbb{E}[R_t]$, which equals the intercept $\bar{Q}$ only if $\mathbb{E}[Q_t] = \mathbb{E}[R_t]$
  - That fails in general: R_t rescales Q_t by its own random diagonal
  - The corrected DCC of Aielli (2013) uses $\mathrm{diag}(Q_{t-\mathbf{1}})^{\mathbf{1}/\mathbf{2}}\varepsilon_{t-\mathbf{1}}$ in place of $\varepsilon_{t-\mathbf{1}}$, so that $\mathbb{E}[Q_t] = \bar{Q}$ and targeting remains consistent
- Step-2 standard errors treat the GARCH estimates and $\bar{Q}$ as known: propagate their error with the stacked sandwich of Newey & McFadden (1994) or by bootstrapping both steps (▸ Appendix)
- CCC against DCC: $H_0 : a = 0$
- b is not identified under H_0, so the χ^2 reference is only indicative: bootstrap under CCC

DCC Estimates for SPY and TLT

Step 1: GARCH(1,1), returns in %

	ω	α	β	$\alpha+\beta$
SPY	0.0283	0.127	0.848	0.975
TLT	0.0057	0.049	0.943	0.993

Step 2: DCC(1,1) and three standard errors

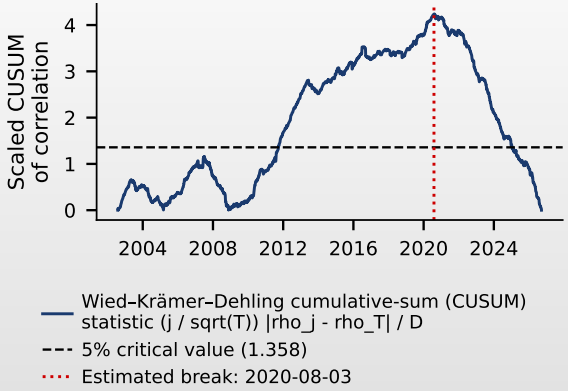
	estimate	Hessian	boot. step 2	boot. both
a	0.0522	0.0059	0.0051	0.0052
b	0.9335	0.0084	0.0077	0.0077
$a+b$	0.9857	–	0.0038	0.0040

- Daily data 31 July 2002 – 18 September 2026, $T = 6\,073$
- Target $\bar{Q}_{\mathbf{12}} = -0.248$
- Persistence $a + b = 0.986$: a shock to Q_t halves in $\ln 0.5 / \ln 0.986 \approx 48$ days
- For the correlation R_t this half-life is only approximate: the recursion is linear in Q_t, not in R_t
- Likelihood ratio (LR) against CCC 533.4
- Bootstrap of both steps under CCC ($B = 199$): largest null value 15.7, $p = 0.005$: CCC rejected
- Step 1 raises the se by only 2% (a) and 1% (b), because T is large
- The Hessian se($\hat{b}$) is conservative here
- TLT volatility is more persistent (larger β) but reacts less to news (smaller α) than SPY volatility

Testing Constant Correlation at an Unknown Date

- Tse (2000): a Lagrange multiplier test of constant conditional correlation, computed from the CCC fit alone
- Picking a break date from a chart and then testing it with a fixed-date Fisher z test is oversized: the date is itself a function of the data
- Wied, Krämer & Dehling (2012): $Q_T = \max_{2 \le j \le T} \frac{j}{\sqrt{T}\,\hat{D}} \left|\hat{\rho}_j - \hat{\rho}_T\right|$, $\hat{\rho}_j$ the correlation of the first j observations
  - Compares the correlation of the first j days with the full-sample $\hat{\rho}_T$
  - A large gap at some j signals a break
  - $\hat{D}^2$: the delta-method long-run variance of $\hat{\rho}$, Bartlett kernel with bandwidth $\lfloor T^{1/4} \rfloor$ ($\lfloor \cdot \rfloor$: integer part)
  - Under H_0 (constant correlation, means and variances): $Q_T \to \sup_{0 \le s \le 1} |B(s)|$, B a Brownian bridge (a random walk in continuous time tied to 0 at $s = 0$ and $s = 1$), $s = j/T$
  - 5% critical value 1.358
  - Constant variances are required, so we apply it to GARCH-standardised residuals
  - Break date estimate $\hat{j} = \arg\max$

Dating the SPY–TLT Correlation Break (1/2)



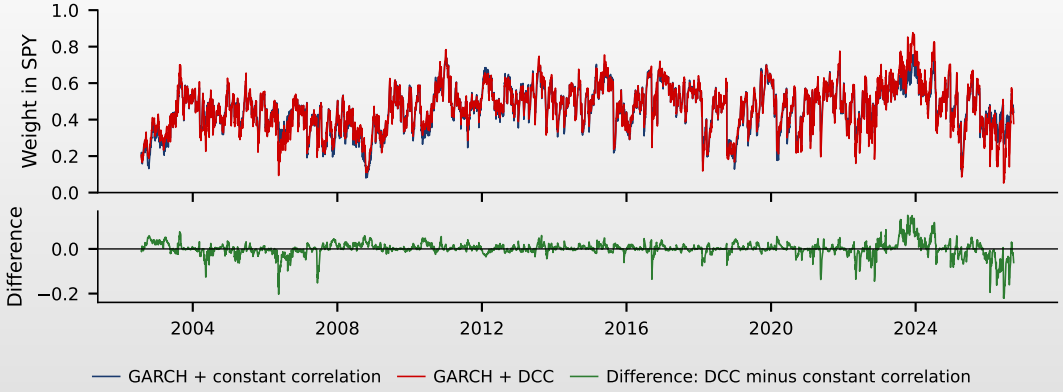
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Dating the SPY–TLT Correlation Break (2/2)

- Wied–Krämer–Dehling process on the GARCH(1,1)-standardised residuals, $T = 6\,073$, bandwidth 8
- Dashed: 5% critical value
- $Q_T = 4.25$ against 1.358: constant correlation rejected ($p < 0.001$)
- Estimated break 3 August 2020, not January 2022: residual correlation -0.36 before, 0.07 after
- Binary segmentation (the test repeated on each sub-period): before the main break, $Q = 3.01$ and a second break on 9 July 2007
- After the main break (August 2020 onwards): $Q = 1.56 > 1.358$, $p = 0.015$, so the correlation is not constant there either
- The limit law is proved for observed series: on estimated residuals the p-values are approximate
- One DCC target over 2002–2026 averages at least two regimes

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Application: Minimum-Variance Weights with DCC (1/2)



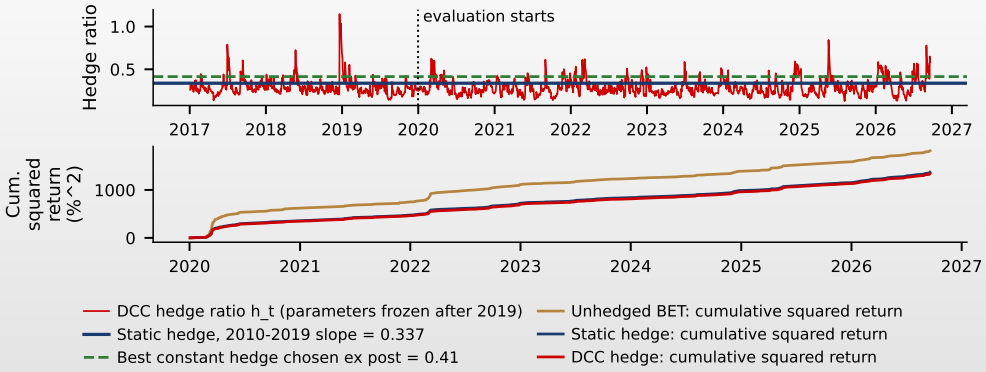
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Application: Minimum-Variance Weights with DCC (2/2)

- Weight in SPY of the daily minimum-variance SPY–TLT portfolio, $w_t = (\sigma^2_{2t} - \sigma_{12,t})/(\sigma^2_{1t} + \sigma^2_{2t} - 2\sigma_{12,t})$, truncated to $[0, 1]$, $\sigma_{12,t} = \rho_t \sigma_{1t} \sigma_{2t}$ the conditional covariance
- Same GARCH volatilities, correlation from DCC or constant, equal to the full-sample 2002–2026 return correlation (-0.30)
- Annualised volatility: 60/40 fixed 10.99%, GARCH with constant correlation 9.10%, GARCH with DCC 9.06%
- Most of the gain comes from GARCH volatilities
- DCC adds little. Mean absolute weight gap DCC minus constant 0.019, largest 0.22 (8 June 2026)
- In-sample illustration: a fair test re-estimates on expanding windows and adds transaction costs (transaction costs: Chapter 4; out-of-sample evaluation: Chapter 8)

MFM_ch6_dcc_minvar

Dynamic Hedging Out of Sample: BET with Euro Stoxx 50 (1/2)



MFM_ch6_corr_inference

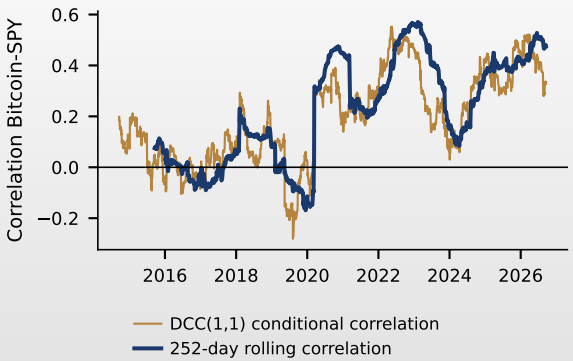
Deeper study

Dynamic Hedging Out of Sample: BET with Euro Stoxx 50 (2/2)

- Daily BET and Euro Stoxx 50 (almost synchronous closes) from 5 January 2010
- Estimation to end-2019 ($n = 2\,491$), evaluation 2020–2026 ($n = 1\,663$) with frozen parameters
- Static hedge: slope in the estimation window, $h = 0.337$
- DCC hedge $h_t = \rho_t \sigma_{S,t} / \sigma_{F,t}$ (S: BET, F: Euro Stoxx 50; $\hat{a} = 0.009$, $\hat{b} = 0.964$): mean 0.29, range 0.13 to 0.84
- Variance removed in 2020–2026: static 24.7%, DCC 26.2%
- The best constant hedge chosen ex post ($h = 0.41$): 25.6%
- Diebold & Mariano (1995) test: a t-test on the mean of the daily loss difference d_t with a HAC se, here d_t = squared hedged return, static minus DCC. Result: $t = 1.29$, one-sided $p = 0.098$, a small gain, not significant at 5%

MFM_ch6_corr_inference

Bitcoin and US Equities: DCC Correlation



- Bitcoin and SPY on common trading days since 18 September 2014 ($T = 3\,018$)
- Weekend Bitcoin moves enter the Monday return
- Average DCC correlation 0.04 before 2020, 0.31 since January 2020
- Peak 0.55 on 19 May 2022
- Very persistent: $a = 0.015$, $b = 0.982$, $a + b = 0.9968$
- LR 105.6, bootstrap p under CCC $= 0.005$: CCC rejected
- From diversifier to risk asset: the full-sample correlation (0.24) hides two regimes
- Pele et al. (2023) ask whether cryptos have become an asset class of their own

MFM_ch6_dcc_btc_spy

Contagion or Interdependence?



Lehman Brothers headquarters, New York, 15 September 2008, the day of the bankruptcy filing

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons

- Common finding: correlations measured during crises are higher than in calm periods
- A high crisis correlation is not yet contagion, it must exceed the calm one
- Two very different explanations
  - **Contagion**: the transmission mechanism itself changes
  - **Volatility bias**: the mechanism is unchanged, but a more volatile source market mechanically raises the correlation
- Forbes & Rigobon (2002): after correcting the bias, most of the 1987, 1994 and 1997 "contagion" became interdependence
- Longin & Solnik (2001), Ang & Chen (2002): correlations rise in bear markets, not in bull markets, an asymmetry the Normal distribution cannot produce

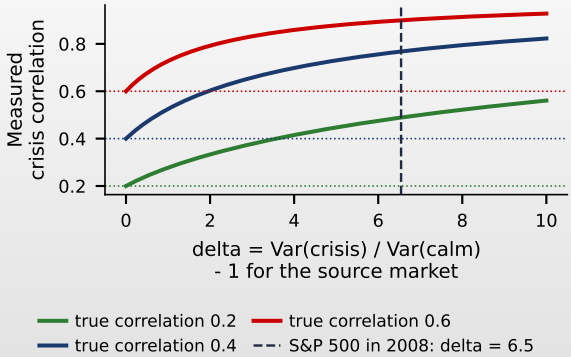
The Heteroskedasticity Bias (1/2): Correlation in a Calm Period

- Model: $y_t = \alpha + \beta x_t + \varepsilon_t$, $\mathrm{Cov}(x_t, \varepsilon_t) = 0$, β and σ_ε^2 constant
  - x_t: return of the source market
  - y_t: return of the other market
  - β: transmission coefficient
  - ε_t: own shock of y, variance σ_ε^2
  - σ_x^2: variance of x
- Two moments of the model
  - $\mathrm{Cov}(x_t, y_t) = \beta\sigma_x^2$ and $\mathrm{Var}(y_t) = \beta^2\sigma_x^2 + \sigma_\varepsilon^2$, because ε_t is uncorrelated with x_t
- Correlation: $\rho^2 = \dfrac{\beta^2\sigma_x^2}{\beta^2\sigma_x^2 + \sigma_\varepsilon^2} = \left(1 + \dfrac{\sigma_\varepsilon^2}{\beta^2\sigma_x^2}\right)^{-1}$
- The same in one number: the signal-to-noise ratio $k = \beta^2\sigma_x^2/\sigma_\varepsilon^2$
  - $\rho^2 = \dfrac{k}{1+k}$, equivalently $k = \dfrac{\rho^2}{1-\rho^2}$: the correlation depends on σ_x^2 only through k

The Heteroskedasticity Bias (2/2): From Calm to Crisis

- Crisis (subscript c): $\sigma^2_{x,c} = (1+\delta)\,\sigma^2_x$, $\delta > 0$ the relative rise of the source variance
- Nothing else changes
- Only σ^2_x grows, so $k_c = (1+\delta)\,k$ and
  - $\rho_c^2 = \dfrac{k_c}{1+k_c} = \dfrac{(1+\delta)\rho^2/(1-\rho^2)}{1+(1+\delta)\rho^2/(1-\rho^2)} = \dfrac{(1+\delta)\rho^2}{1+\delta\rho^2}$
- Then $\rho_c = \rho\sqrt{\dfrac{1+\delta}{1+\delta\rho^2}}$, so $|\rho_c| > |\rho|$ for $0 < |\rho| < 1$: the measured correlation rises in absolute value although β is the same
- Inverting: $\rho^* = \dfrac{\rho_c}{\sqrt{1+\delta(1-\rho_c^2)}}$ is the crisis correlation adjusted for the volatility increase
- Quick check: the source variance doubles ($\delta = 1$) and $\rho = 0.5$
  - $k = 0.25/0.75 = 1/3$, $k_c = 2/3$, $\rho_c^2 = (2/3)/(5/3) = 0.4$, so $\rho_c = 0.632$: a rise of 0.13 with no change in β
  - Adjusting: $\rho^* = 0.632/\sqrt{1+1\cdot(1-0.4)} = 0.632/1.265 = 0.5$, the calm value is recovered

How Much Can Volatility Alone Raise a Correlation?



- Measured crisis correlation ρ_c as a function of δ, for three true correlations that do not change
- Vertical line: the variance of 2-day S&P 500 returns in the 2008 crisis window was 7.5 times its calm-year level ($\delta = 6.54$)
- At this δ an unchanged true correlation of 0.4 is measured as 0.77, and 0.2 as 0.49

MFM_ch6_fr_bias

The Forbes–Rigobon Test and Its Inference

- Procedure
  - Choose the source market x, a calm window and a crisis window
  - Use 2-day returns to neutralise the different closing times
  - Compute $\delta = \hat{\sigma}^2_{x,c}/\hat{\sigma}^2_{x,calm} - 1$ and ρ^*
  - Test $H_0 : \rho^* \leq \rho_{calm}$ with Fisher z statistics, the convention of Forbes & Rigobon (2002)
- Inference
  - With δ fixed, $\mathrm{Var}(\operatorname{artanh}\rho^*) \approx 1/[n_c(1+\delta(1-\rho_c^2))] < 1/n_c$ (n_c: crisis observations): the Fisher test is conservative
  - δ is estimated and 2-day returns overlap: resample the calm and crisis windows jointly (stationary bootstrap, mean block 10 days, $B = 1\,999$ draws) and test $D = \rho^* - \rho_{calm}$
  - Stationary bootstrap: it resamples blocks of consecutive days with random starts and geometric random lengths, which keeps the serial dependence inside each block

Assumptions to remember

- No omitted common shock: a global factor hitting both markets raises σ_ε as well as σ_x, and the adjustment over-corrects (Corsetti, Pericoli & Sbracia (2005); Appendix)
- No feedback from y to x (no endogeneity)
- The choice of windows matters: results can flip with a few weeks
- The correction goes too far if the mechanism really did change

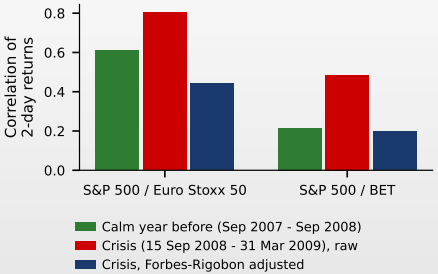
Worked Example: The 2008 Crisis Step by Step (1/2): Adjusting

- Source market: S&P 500
- Calm year: 14 September 2007 – 12 September 2008 ($n = 242$ 2-day returns)
- Crisis: 15 September 2008 – 31 March 2009 ($n = 130$)
- Variance ratio: $1 + \delta = 7.54$, so $\delta = 6.54$
- Euro Stoxx 50: $\rho_{calm} = 0.610$, $\rho_c = 0.804$
  - $\rho^* = 0.804/\sqrt{1 + 6.54\,(1 - 0.804^2)} = 0.442$
- BET: $\rho_{calm} = 0.216$, $\rho_c = 0.485$
  - $\rho^* = 0.485/\sqrt{1 + 6.54\,(1 - 0.485^2)} = 0.198$
- Both raw crisis correlations exceed the calm ones by far
- After the adjustment both fall below them

Worked Example: The 2008 Crisis Step by Step (2/2): Testing

- One-sided Fisher test of $H_0 : \rho_1 \leq \rho_{calm}$, with $\rho_1 = \rho_c$ (raw) or $\rho_1 = \rho^*$ (adjusted)
  - $z = \dfrac{\operatorname{artanh}\rho_1 - \operatorname{artanh}\rho_{calm}}{\sqrt{1/(n_c - 3) + 1/(n_{calm} - 3)}}, \quad p = 1 - \Phi(z)$
- Sample sizes: consecutive 2-day returns share one day, so they are not independent
  - We use effective sizes $n_{calm} = 121$ and $n_c = 65$, half of 242 and 130
  - For i.i.d. Normal daily pairs, the Bartlett approximation gives an effective size near $2n/3$, so halving is a cautious convention in that benchmark, not a general bound
- Euro Stoxx 50, raw: $z = \dfrac{\operatorname{artanh} 0.804 - \operatorname{artanh} 0.610}{\sqrt{1/62 + 1/118}} = 2.56$
  - One-sided p-value: 0.005 with ρ_c, 0.932 with ρ^*
- BET
  - One-sided p-value: 0.024 with ρ_c, 0.548 with ρ^*
- Check: with all 130 and 242 overlapping returns the raw Euro Stoxx 50 statistic from unrounded correlations would be $z = 3.66$, $p \approx 0.0001$: treating overlapping returns as independent overstates the evidence

2008: Raw versus Adjusted Crisis Correlations



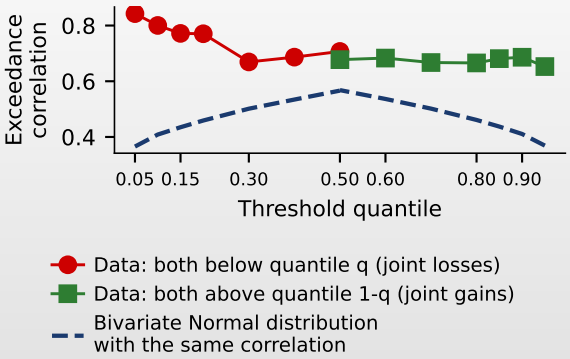
- Correlation of 2-day returns with the S&P 500: calm year, crisis (raw) and after the Forbes–Rigobon adjustment
- Both rises significant at 5%, neither survives the adjustment
- With δ estimated: $\hat{\delta} = 6.54$, 90% interval [3.9, 9.6]
- BET: 90% interval for ρ^* [0.16, 0.24], $p = 0.62$
- Euro Stoxx 50: 90% interval for ρ^* [0.38, 0.52], $p = 0.98$
- The block bootstrap uses all return pairs and the covariance between the calm-window and crisis-window estimates
- For BET the bootstrap gives a smaller standard error: 0.025, against 0.047 from Normal theory
- Verdict: no evidence of an increase once volatility is adjusted for, under the maintained assumptions
- The global 2008 shock challenges the assumptions
- The test has low power (the 2020 application: Seminar B3)

MFM_ch6_forbes_rigobon

Exceedance Correlations

- Longin & Solnik (2001): correlation conditional on both returns being beyond a threshold
- Lower tail at quantile q: $\rho^{-}(q) = \mathrm{Corr}\big(x, y \mid x \le F_x^{-1}(q),\, y \le F_y^{-1}(q)\big)$
- $F_x^{-1}(q)$: the q-quantile of x (e.g. $q = 0.10$: the worst 10% of weeks)
- Upper tail: the same with $x \ge F_x^{-1}(1-q)$, $y \ge F_y^{-1}(1-q)$
- Benchmark: for a bivariate Normal distribution, $\rho^{\pm}(q) \to 0$ as $q \to 0$
  - Conditioning on extremes lowers the correlation even when nothing changes: compare with the Normal curve, not with ρ

Exceedance Correlations: S&P 500 and Euro Stoxx 50



- Weekly returns 2000–2026 on common S&P 500–Euro Stoxx 50 days, without BET ($n = 1\,393$, overall $\hat{\rho} = 0.78$)
- Dashed: the bivariate Normal distribution with the same correlation (2 million simulated pairs)
- At $q = 0.10$: joint losses $\rho^{-} = 0.80$ (86 weeks), joint gains $\rho^{+} = 0.69$ (70 weeks)
- Normal benchmark 0.41
- Both tails lie far above the Normal benchmark, and crashes are shared more than rallies
- A single coefficient cannot describe this: we need a model of the whole dependence structure, a copula

MFM_ch6_exceedance

Why Correlation Is Not Enough

- Embrechts, McNeil & Straumann (2002): three fallacies about linear correlation
- Correlation measures only *linear* dependence and is not invariant to monotone transformations: $\text{Corr}(x, y) \neq \text{Corr}(e^x, e^y)$
- Margins and correlation do not determine the joint distribution
  - Two portfolios with identical margins and ρ can have very different Value-at-Risk (Chapter 7)
- Attainable correlations: for given margins, $\rho \in [\rho_{\min}, \rho_{\max}]$, reached by countermonotone and comonotone pairs
  - Lognormal $X = e^{\sigma_1 Z}$, $Y = e^{\pm\sigma_2 Z}$, Z standard Normal ($+$: comonotone, $-$: countermonotone): $\rho_{\max} = \dfrac{e^{\sigma_1\sigma_2} - 1}{\sqrt{(e^{\sigma_1^2} - 1)(e^{\sigma_2^2} - 1)}}$, $\rho_{\min}$ with $e^{-\sigma_1\sigma_2}$ in the numerator
  - $\sigma_1 = 1$, $\sigma_2 = 3$: $\rho_{\max} = 0.162$ and $\rho_{\min} = -0.008$: perfectly dependent, yet almost uncorrelated

Sklar's Theorem: Margins plus Copula

- A **copula** is a joint distribution function on $[0,1]^d$ with uniform margins
- Sklar (1959): for any joint distribution F with margins $F_1, \ldots, F_d$ there is a copula C with $F(x_1, \ldots, x_d) = C\big(F_1(x_1), \ldots, F_d(x_d)\big)$
  - d: number of variables
  - F_i: marginal distribution function of x_i
  - $u_i = F_i(x_i) \in [0,1]$: its probability level
  - C is unique if the margins are continuous
  - Conversely, any copula and any margins give a valid joint distribution
  - With densities: $f(x) = c\big(F_1(x_1), \ldots, F_d(x_d)\big) \prod_i f_i(x_i)$, c the copula density, f_i the marginal densities (the comonotone copula has none)
- Margins carry volatility, skewness and fat tails of each asset (Chapters 1 and 5)
- The copula carries all the dependence and is invariant to increasing transformations of each variable
- Three reference copulas and the Fréchet–Hoeffding bounds $W \le C \le M$ (Nelsen (2006))
  - Independence $\Pi(u,v) = uv$
  - Comonotonicity $M(u,v) = \min(u,v)$
  - Countermonotonicity $W(u,v) = \max(u+v-1, 0)$

Rank Correlations: Kendall and Spearman

- **Kendall's tau** (Kendall (1938)): concordant minus discordant pairs
  - $\tau = P\big[(X_1 - X_2)(Y_1 - Y_2) > 0\big] - P\big[(X_1 - X_2)(Y_1 - Y_2) < 0\big]$
  - (X_1, Y_1), (X_2, Y_2): two independent draws of the pair
  - Concordant: both coordinates move in the same direction
  - Copula form: $\tau = 4 \int\int C \, dC - 1 = 4\,\mathbb{E}[C(U, V)] - 1$
- **Spearman's rho** (Spearman (1904)): Pearson correlation of the ranks
  - Copula form: $\rho_S = 12 \int\int (C(u, v) - uv) \, du \, dv$
- Both lie in $[-1, 1]$ and depend only on the copula: 1 for M, 0 for Π, -1 for W
- θ: the copula parameter

Links to copula parameters

Copula	τ
Gaussian, t	$\frac{2}{\pi} \arcsin \rho$
Clayton	$\theta/(\theta + 2)$
Gumbel	$1 - 1/\theta$
Frank	$1 - \frac{4}{\theta}\left[1 - D_1(\theta)\right]$

- $D_1(\theta) = \frac{1}{\theta} \int_0^\theta \frac{t}{e^t - 1} dt$ (Debye function)
- Gaussian: $\rho_S = \frac{6}{\pi} \arcsin(\rho/2)$
- Example, Gaussian copula with $\rho = 0.5$: $\tau = \frac{2}{\pi} \cdot \frac{\pi}{6} = 1/3$ and $\rho_S = \frac{6}{\pi} \arcsin 0.25 = 0.483$

Five Copula Families

Family	$C(u, v)$	Parameter
Gaussian	$\Phi_{\mathbf{2},\rho}\left(\Phi^{-\mathbf{1}}(u), \Phi^{-\mathbf{1}}(v)\right)$	$-1 < \rho < 1$
Student t (Demarta & McNeil (2005))	$t_{\mathbf{2},\rho,\nu}\left(t_\nu^{-\mathbf{1}}(u), t_\nu^{-\mathbf{1}}(v)\right)$	$-1 < \rho < 1$, $\nu > 0$
Clayton (Clayton (1978))	$\left(u^{-\theta} + v^{-\theta} - 1\right)^{-\mathbf{1}/\theta}$	$\theta > 0$
Gumbel (Gumbel (1960))	$\exp\left(-\left[(-\ln u)^\theta + (-\ln v)^\theta\right]^{\mathbf{1}/\theta}\right)$	$\theta \geq 1$
Frank (Frank (1979))	$-\frac{\mathbf{1}}{\theta} \ln\left(1 + \frac{(e^{-\theta u} - \mathbf{1})(e^{-\theta v} - \mathbf{1})}{e^{-\theta} - \mathbf{1}}\right)$	$\theta \neq 0$

- Φ, $\Phi^{-\mathbf{1}}$: standard Normal distribution and quantile functions
- $\Phi_{\mathbf{2},\rho}$: bivariate Normal distribution function with correlation ρ
- t_ν, $t_{\mathbf{2},\rho,\nu}$: the Student t analogues, ν degrees of freedom
- θ: dependence parameter
- Elliptical (Gaussian, t): extend to d dimensions with a correlation matrix, the standard choice in risk systems
- $\nu \to \infty$ turns the t copula into the Gaussian one
- Archimedean (Clayton, Gumbel, Frank): $C(u, v) = \varphi^{-\mathbf{1}}\left(\varphi(u) + \varphi(v)\right)$, φ: generator (decreasing, $\varphi(1) = 0$)
  - One parameter, closed form, easy simulation, but exchangeable (symmetric in the variables, $C(u, v) = C(v, u)$) and poor in high dimension
- Rotation by 180° (the survival copula of $(1 - U, 1 - V)$) moves the tail dependence to the opposite corner (densities and generators in the Appendix)

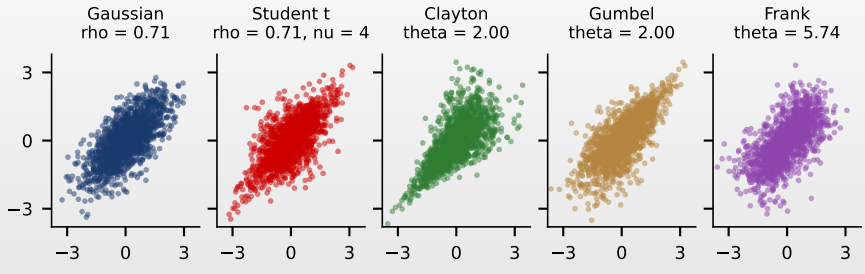
Tail Dependence

- Lower tail:
  $\lambda_L = \lim_{q \downarrow \mathbf{0}} P\big(V \le q \mid U \le q\big) = \lim_{q \downarrow \mathbf{0}} C(q,q)/q$, with
  $U = F_{\mathbf{1}}(X_{\mathbf{1}})$, $V = F_{\mathbf{2}}(X_{\mathbf{2}})$
- Upper tail: $\lambda_U = \lim_{q \downarrow \mathbf{0}} P\big(V > 1 - q \mid U > 1 - q\big)$
- q: tail probability (threshold level)
- $\lambda_L, \lambda_U \in [0,1]$
- $t_{\nu+\mathbf{1}}$ in the table: Student t distribution function with $\nu + 1$ degrees of freedom
- $\lambda > 0$: the probability of a joint extreme stays positive however extreme the threshold
- $\lambda = 0$: asymptotic independence
- Tail dependence is a copula property, invisible in ρ
- At a finite q, $C(q,q)/q$ is a probability, not yet the limit

Closed forms

	λ_L	λ_U
Gaussian	0	0
Student t	$2\, t_{\nu+\mathbf{1}}\Big(-\sqrt{\frac{(\nu+\mathbf{1})(\mathbf{1}-\rho)}{\mathbf{1}+\rho}}\Big)$	same
Clayton	$2^{-\mathbf{1}/\theta}$	0
Gumbel	0	$2 - 2^{\mathbf{1}/\theta}$
Frank	0	0

Five Copulas with the Same Kendall's Tau



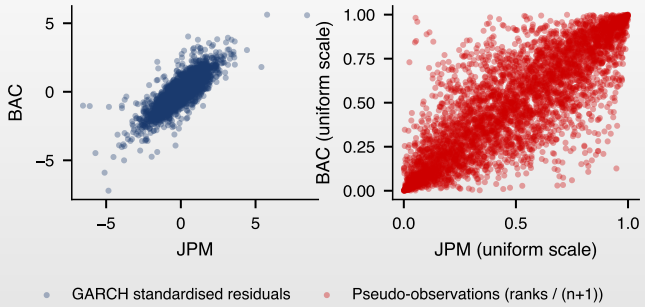
- 1 500 simulated pairs from each copula with $\tau = 0.5$ on Normal-score margins $(\Phi^{-1}(u), \Phi^{-1}(v))$
- Gaussian and t $\rho = 0.707$ (t with $\nu = 4$), Clayton and Gumbel $\theta = 2.0$, Frank $\theta = 5.74$
- Corners: Gaussian and Frank thin out, t stays dense in both, Clayton is sharp lower-left (joint crashes), Gumbel sharp upper-right
- Equal Kendall dependence does not determine joint-tail risk

MFM_ch6_copula_zoo

Estimating Copulas

- **Full maximum likelihood**: margins and copula estimated jointly
  - Efficient, but computationally demanding and sensitive to misspecification
- **IFM** (inference functions for margins, Joe (2005)): two steps
  - Fit parametric margins (e.g. GARCH with Student t)
  - Set $\hat{u}_{it} = \hat{F}_i(x_{it})$ ($\hat{F}_i$: fitted marginal distribution function)
  - Maximise $\sum_t \ln c(\hat{u}_{1t}, \hat{u}_{2t}; \theta)$, c the copula density
- **CML** (canonical maximum likelihood, Genest, Ghoudi & Rivest (1995)): no parametric margins
  - Pseudo-observations $\hat{u}_{it} = \text{rank}(x_{it})/(T+1)$
  - Then maximise the copula likelihood
  - Robust to misspecified margins
  - The $T+1$ keeps $\hat{u}$ away from 0 and 1
- For returns: filter each series with GARCH first, then use the ranks of the standardised residuals
- Model choice: AIC $= 2k - 2\ell$ (Akaike information criterion), BIC $= k \ln T - 2\ell$ (Bayesian information criterion)
- k: number of parameters, ℓ: maximised log-likelihood
- The lower, the better

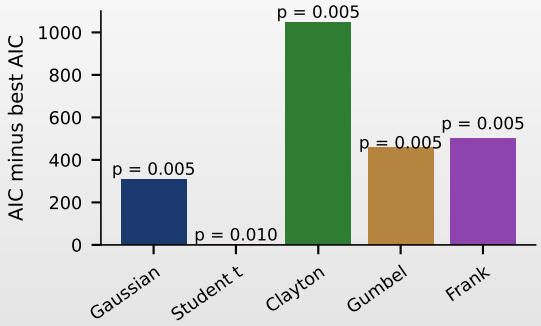
From Residuals to Pseudo-Observations: JPM and BAC



- JPMorgan Chase (JPM) and Bank of America (BAC), daily 2010–2026 ($T = 4\,202$), GARCH(1,1)-filtered
- Kendall's $\hat{\tau} = 0.641$, Spearman's $\hat{\rho}_S = 0.822$
- Both corners of the uniform scatter are dense: joint crashes and joint rallies

MFM_ch6_copula_banks

Copula Estimates and Goodness of Fit: JPM and BAC



Numbers above bars: goodness-of-fit p-value
(Rosenblatt Cramer-von Mises, parametric bootstrap)

	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	λ_L	λ_U
Gaussian	0.833	–	0.000	0.000
Student t	0.842	4.24	0.531	0.531
Clayton	2.384	–	0.748	0.000
Gumbel	2.587	–	0.000	0.693
Frank	9.152	–	0.000	0.000

- The bars show the AIC distance, the numbers above them the bootstrap goodness-of-fit p-values (full table in the Appendix)
- Student t wins by far: $\hat{\nu} = 4.2$, $\hat{\lambda} = 0.53$ in both tails
- Gaussian beats the Archimedean ones: strong, roughly symmetric dependence
- The Rosenblatt test (slide "Goodness of Fit") rejects every family at 5%
- The t copula has the smallest statistic: 0.082, against 0.968 for the Gaussian

MFM_ch6_copula_banks

Deeper study

Inference for Copula Parameters (1/2): Inverting Kendall's Tau

- **Inversion of** $\hat{\tau}$: a U-statistic with a Hoeffding-projection standard error
- $\hat{\theta} = g(\hat{\tau})$ by the delta method (Genest & Rivest (1993))
  - g: the inverse of the link $\tau(\theta)$ of the previous table
  - U-statistic: an average over all pairs of observations
  - Elliptical copulas: $\tau = \frac{2}{\pi} \arcsin \rho$ for *every* ν, so τ identifies ρ, never ν
  - JPM–BAC: $\hat{\tau} = 0.641$ (se 0.0065), $\hat{\rho}_\tau = \sin(\pi\hat{\tau}/2) = 0.845$ (se 0.0055)
  - CML $\hat{\rho} = 0.842$ (se 0.0053)

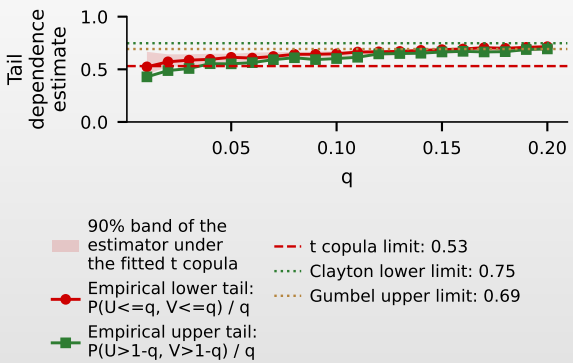
Inference for Copula Parameters (2/2): Ranks, Fit and GARCH Filtering

- **CML: the ranks are estimated**, so the inverse Hessian is too small
- The Genest, Ghoudi & Rivest (1995) sandwich adds the rank terms
  - Increase of the se: +60% for the Gaussian ρ (S&P 500–Euro Stoxx 50 weekly, 0.00863 to 0.01380), +17% for the t-copula ρ (0.004576 to 0.005334) and +1% for ν (JPM–BAC)
- Not all of the +60% is the price of ranking
  - If the Gaussian copula were correct, estimating the margins by ranks would multiply the se by $\sqrt{1+\rho^2}$ (Appendix)
  - At the weekly correlation 0.78 the factor would be 1.27; Monte Carlo at $\rho = 0.8$ gives 1.28
  - The observed factor, 1.60, points to a misspecified Gaussian copula
  - The sandwich stays valid in this case too (fit: Seminar B2)
- **After GARCH filtering** (Chen & Fan (2006)): CML on the ranks of estimated residuals has the same limit as on the true innovations
  - Conditions: i.i.d. innovations with copula C_θ and a correctly specified GARCH
  - Then the sandwich stays valid and, unlike DCC, no step-1 correction is needed (Monte Carlo in the Appendix)

Goodness of Fit: The Rosenblatt Test

- If $(U, V) \sim C_\theta$, then $E_1 = U$ and $E_2 = h(V \mid U) = \partial C_\theta(U, V)/\partial u$ (the conditional distribution function of V given U) are independent uniforms
- Statistic (Genest, Rémillard & Beaudoin (2009)): $S_n = n \int_{[0,1]^2} \left(D_n(e) - e_1 e_2\right)^2 de$
  - $e = (e_1, e_2) \in [0, 1]^2$
  - $D_n(e)$: share of the n pairs with $E_1 \leq e_1$ and $E_2 \leq e_2$
  - $e_1 e_2$: the same share under independence
  - $S_n \geq 0$, large values mean a poor fit
- p-value by parametric bootstrap, because θ is estimated
  - Simulate n pairs from $C_{\hat{\theta}}$, re-rank them into pseudo-observations, re-estimate, recompute $S_n^{(b)}$
  - $p = (1 + \#\{S_n^{(b)} \geq S_n\})/(B + 1)$
  - For JPM–BAC $B = 200$ (100 for Student t)
- On GARCH residuals, Rémillard (2017) shows that rank-based tests keep their limit and the parametric bootstrap applies
- With $T = 4\,202$ the test has high power against small departures: a rejection does not say which feature is missing

Empirical Tail Dependence: JPM and BAC



- $\hat{\lambda}_L(q) = \hat{C}(q,q)/q$ and $\hat{\lambda}_U(q)$ for q from 1% to 20%
- Band: 90% range of the same estimator on data simulated from the fitted t copula
- $q = 5\%$: $\hat{\lambda}_L = 0.61$, $\hat{\lambda}_U = 0.55$
- $q = 1\%$: 0.52 and 0.43
- When JPM has one of its 5% worst days, BAC is in its own worst 5% about 61% of the time
- Lower tail inside the t-copula band (0.57–0.65 at 5%): the t copula describes the crashes well
- Finite-q values are threshold probabilities, not the limit λ
- Few points lie in the far corner

MFM_ch6_copula_banks

Deeper study

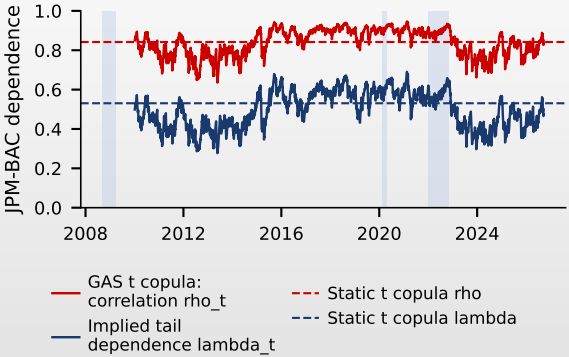
Is the Tail Asymmetry Real?

- Schmidt & Stadtmüller (2006): $\hat{\lambda}_L = \frac{n}{k}\,\hat{C}_n\left(\frac{k}{n}, \frac{k}{n}\right)$, the chart estimator with $q = k/n$
  - n: sample size
  - k: number of observations in the tail
  - $\hat{C}_n$: empirical copula
  - A larger k lowers the variance but raises the bias (conditions in the Appendix)
- $H_0 : \lambda_L(q) = \lambda_U(q)$ at a fixed q (threshold probabilities, not the limits)
- Bootstrap of residual pairs, re-ranked, $B = 999$

q	k	$\hat{\lambda}_L - \hat{\lambda}_U$	95% interval, i.i.d.	95% interval, blocks
5%	210	0.062	[−0.014, 0.148]	[−0.014, 0.138]
1%	42	0.095	[−0.095, 0.286]	[−0.119, 0.286]

- The block version (stationary bootstrap, mean block 1.6 and 1.0 days, Politis & White (2004) rule) allows serial dependence in the tail indicators
- The interval at 1% is wider, as the bias–variance trade-off predicts
- All intervals contain zero: the asymmetry is not significant, so the symmetric t copula is not contradicted in the tails

A Dynamic t Copula for JPM and BAC (1/2)



MFM_ch6_copula_inference

Deeper study

A Dynamic t Copula for JPM and BAC (2/2)

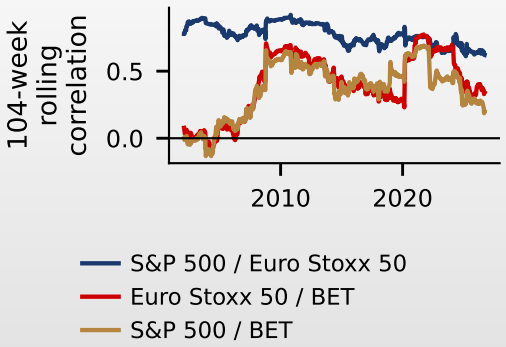
- GAS (generalised autoregressive score) t copula, Creal, Koopman & Lucas (2013): $\rho_t = \tanh f_t$, $f_{t+1} = \omega + A\, s_t + B f_t$, $s_t = \partial \ln c(\hat{u}_t, \hat{v}_t; \rho_t, \nu)/\partial f_t$ (unit scaling), same pseudo-observations
- A: reaction to the score s_t, B: persistence
- $A = 0.025$, $B = 0.9926$, $\nu = 5.35$ (static 4.24)
- Likelihood ratio against the static t copula 245.5
- ρ_t from 0.64 (23 May 2013) to 0.95 (18 February 2021)
- λ_t from 0.28 to 0.69 around a static 0.53
- Once ρ moves, a larger ν suffices: part of the static fat tails was time variation in dependence
- Under $A = 0$, B is not identified, so the χ^2 distribution does not apply to the LR statistic
- The p-value comes from a bootstrap (procedure: Seminar B6); it is not reported here for JPM–BAC
- Time variation is one explanation consistent with the rejection of every static family (Seminar B6)

MFM_ch6_copula_inference

A Cautionary Tale: The Gaussian Copula and 2008

- Li (2000) modelled joint default times with a Gaussian copula
  - One correlation parameter, calibrated to market spreads
  - It priced whole credit portfolios and became the standard for tranches of collateralised debt obligations (CDO)
- Zero asymptotic tail dependence: for the most extreme joint defaults it gives lower probabilities than tail-dependent models calibrated to the same data
- A static correlation calibrated in calm years: exactly the Forbes–Rigobon and DCC lessons of this chapter

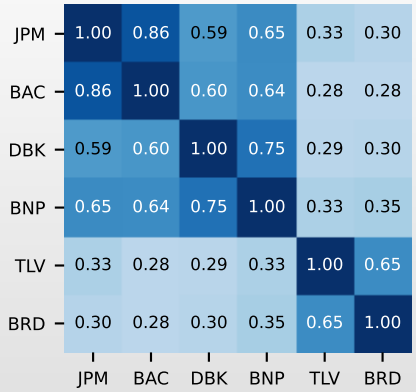
US, Euro-Area and Romanian Equities, 2000–2026

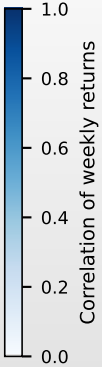


- Weekly returns, 104-week window, common trading days, Friday closes
- Euro Stoxx 50–BET: 0.33 in 2000–2009, 0.45 in 2010–2019, 0.61 since 2020
- S&P 500–Euro Stoxx 50: 0.84, 0.78, 0.70
- Rolling Euro Stoxx 50–BET: 0.31 at end-2007, 0.65 at end-2009, 0.29 at end-2019, 0.76 at end-2021, 0.35 now
- Jumps in crises and partial reversal afterwards: integration or the Forbes–Rigobon volatility bias? (Seminar C2)

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Banks: Correlations of Weekly Returns, 2010–2026 (1/2)





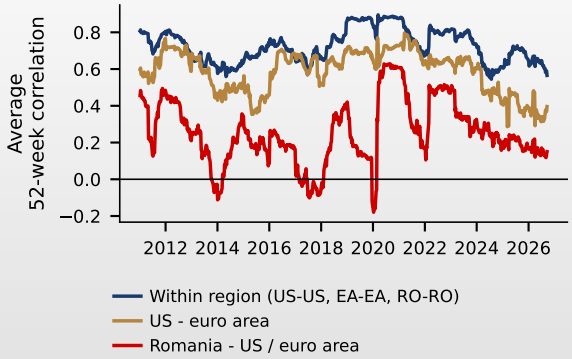
MFM_ch6_banks

Banks: Correlations of Weekly Returns, 2010–2026 (2/2)

- US: JPM and BAC
- Euro area: DBK (Deutsche Bank) and BNP (BNP Paribas)
- Romania: TLV (Banca Transilvania) and BRD (BRD-Groupe Société Générale)
- 871 weeks from 15 January 2010
- Three regional blocks: US 0.86, euro area 0.75, Romania 0.65
- Cross-region links are weakest for Romania: BNP–TLV 0.33, JPM–BRD 0.30
- Bank returns share country-level factors: sovereign risk, domestic rates, regulation

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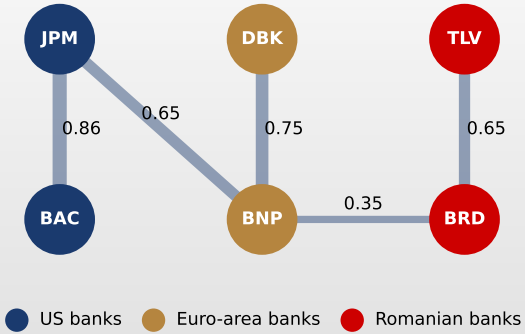
Bank Correlations over Time



- Average pairwise 52-week correlation of weekly returns: within regions (legend: US, EA: euro area, RO: Romania), between the US and the euro area, between Romania and the others
- Full period: within regions 0.73, US–euro area 0.59, Romania–others 0.25
- March–June 2020: 0.87, 0.69 and 0.56: the Romanian link more than doubled in a global shock
- Latest 0.57, 0.40, 0.15
- Diversification across banking systems works in normal times and weakens exactly in systemic events

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A Correlation Network of Banks



- Minimum spanning tree (Mantegna (1999)) on $d_{ij} = \sqrt{2(1-\rho_{ij})}$: the 5 strongest links that connect all 6 banks without cycles
- Edges: JPM–BAC (0.86); JPM–BNP (0.65); DBK–BNP (0.75); BNP–BRD (0.35); TLV–BRD (0.65)
- Romania is attached through BNP–BRD
- Chapter 4 uses $\sqrt{(1-\rho)/2}$: a factor 2 apart, both increase with $1-\rho$, so the tree is the same
- The tree is undirected and contemporaneous: it does not identify transmission paths, only hypotheses for the network measures of Chapter 18 (literature in the Appendix)

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Deeper study

Case Study (1/2): Which Events Shake All Markets at Once?

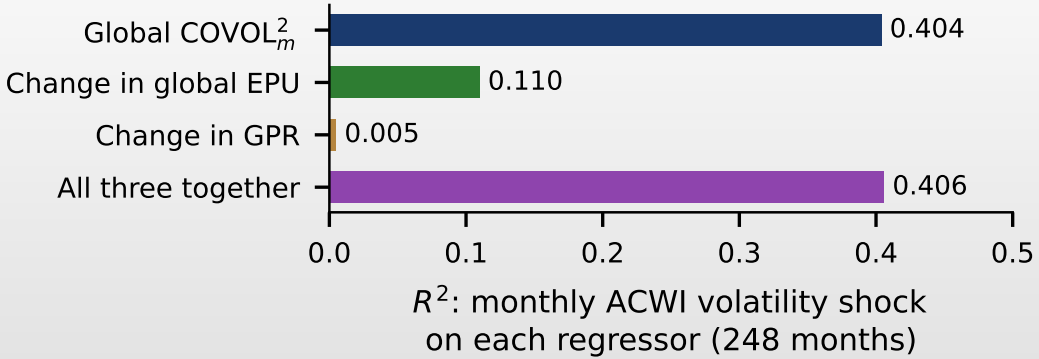
- Questions of Engle & Campos-Martins (2023)
  - which events move the volatilities of almost all countries and asset classes on the same day?
  - how large is this common shock?
- The daily size of the common shock is global COVOL (common volatility)
- The distinct idea: a correct DCC gives residuals e_t with $E_{t-\mathbf{1}}(e_t e_t^\top) = I$, but uncorrelated is not independent: the squares can comove
  - $\Psi_t = E_{t-\mathbf{1}}\left[(e_t^{\mathbf{2}} - \iota)(e_t^{\mathbf{2}} - \iota)^\top\right]$
  - e_t: vector of the N DCC residuals of day t
  - $e_t^{\mathbf{2}}$: its element-by-element square
  - ι: vector of ones
  - $E_{t-\mathbf{1}}$: expectation given the information at $t-1$
  - Off-diagonal elements $\Psi_{ij,t} \neq 0$: the squared shocks of assets i and j are large on the same days
  - The cross-sectional analogue of ARCH (autoregressive conditional heteroskedasticity, Chapter 5: today's variance driven by yesterday's squared shock): correlated squares across assets on one day, not over time
  - Herskovic, Kelly, Lustig & Van Nieuwerburgh (2016) document such a factor in the idiosyncratic volatility of US stocks

Deeper study

Case Study (2/2): One Common Factor Scales Every Variance

- Model: one common factor x_t scales the variance of every residual
  - $e_{it} = \sqrt{s_i x_t + 1 - s_i}\,\varepsilon_{it}$, ε_{it} i.i.d. $\mathcal{N}(0,1)$, $x_t > 0$, $E_{t-\mathbf{1}}(x_t) = 1$
  - x_t: the squared global COVOL of day t ($x_t > 1$: a common shock day)
  - $s_i \in [0,1]$: loading of asset i, the weight of x_t in its conditional variance $g_{it} = s_i x_t + 1 - s_i = 1 + s_i(x_t - 1)$
- Estimation: only the products $s_i(x_t - 1)$ are identified, so the scale must be fixed by a normalisation
  - Equal loadings $s_i = 1$: $\hat{x}_t = \frac{1}{N}\sum_i e_{it}^2$, the cross-sectional mean of the squared residuals, on the variance scale of the residuals. Example: squared residuals $4, 9, 1, 4$ on one day give $\hat{x}_t = 4.5$
  - Unequal loadings: alternating maximum likelihood over x_t (one cross-section per day) and s_i (one time series per asset), with $s^\top s = 1$ and mean $x_t = 1$
  - Under $s^\top s = 1$ the loadings are relative sizes, not variance shares (equal loadings would be $s_i = 1/\sqrt{N}$)
- Test ξ of $H_\mathbf{0}$: $\Psi_{ij} = 0$, $i \neq j$ ($\mathcal{N}(0,1)$ under $H_\mathbf{0}$)
- Data: 47 country ETFs (exchange-traded funds) listed on the NYSE (New York Stock Exchange), 2000–2021, and 23 US asset-class series (details in the Appendix)

Realised Shocks against News Indices (1/2)



MFM_ch6_covol

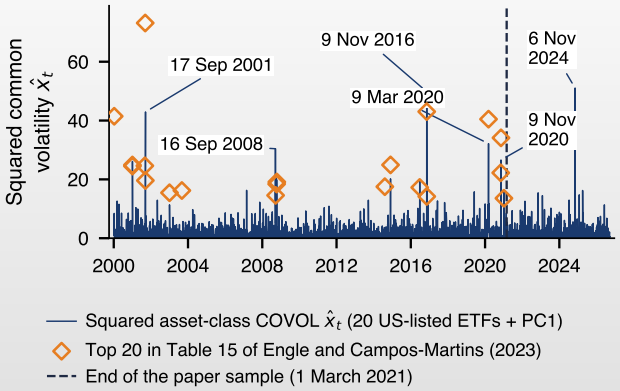
Deeper study

Realised Shocks against News Indices (2/2)

- Monthly, 248 months (Table 8 of Engle & Campos-Martins (2023)): ACWI (MSCI All Country World Index) volatility shock regressed on each index
  - On COVOL_m^2, the monthly mean of $\hat{x}_t$ minus 1: $R^2 = 0.404$ (R^2: share of variance explained)
  - On the change in EPU (economic policy uncertainty, Baker, Bloom & Davis (2016)): 0.110
  - On GPR (geopolitical risk, Caldara & Iacoviello (2022)): 0.005
  - Both news indices added: only 0.406
- Top days in the paper: 9 March 2020 ($\hat{x}_t = 51.4$), 24 June 2016 (Brexit) and 28 April 2020, with a positive mean return (+2.33%)
  - COVOL measures the size of the common shock, not its sign
  - News indices measure what people fear, COVOL the shocks that hit prices (event table in the Appendix)

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Asset-Class COVOL on the Course ETFs, 2000–2026 (1/2)



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Deeper study

Asset-Class COVOL on the Course ETFs, 2000–2026 (2/2)

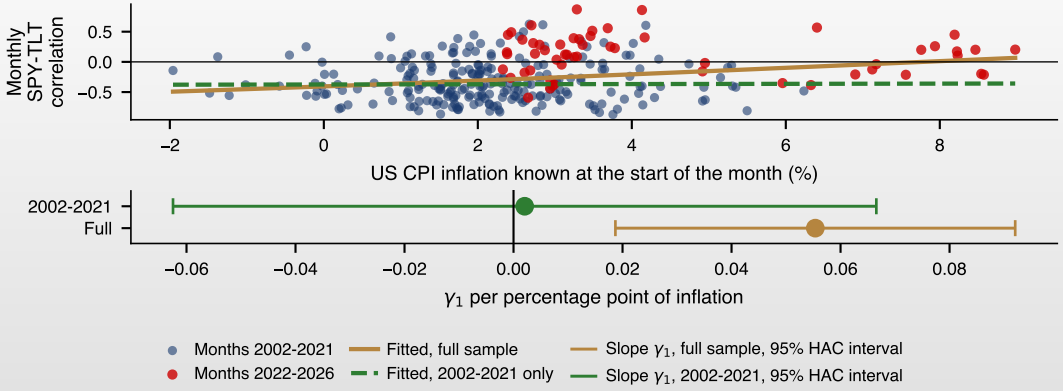
- 20 US-listed ETFs and their PC1 (first principal component), 4 January 2000 – 18 September 2026, 6 717 days
- AR(1) mean (first-order autoregressive) with PC1, GARCH(1,1), then the alternating estimation (13 iterations)
- Diamonds: the 20 largest values of Table 15 of Engle & Campos-Martins (2023)
- Top days up to 1 March 2021: 9 November 2016 ($\hat{x}_t = 49.8$), 17 September 2001 (42.9), 9 March 2020 (32.2): the 2016 US election, the reopening after 9/11 and the COVID-19 crash
- 12 of the 20 largest days of our replication before March 2021 are among the 20 dates of Table 15
- With $s^\top s = 1$ and mean $\hat{x}_t = 1$ the scale depends on the panel
- Replication values are not comparable with the paper's: compare rankings, not levels
- Largest in the full sample: 6 November 2024 (51.1, mean ETF return 0.95%), the day after the 2024 US election: a large common shock on an up day

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An Open Question: Will the Stock–Bond Correlation Stay Positive?

- Questions
  - is the positive stock–bond correlation since 2022 a durable regime governed by inflation?
  - can it be forecast?
- Campbell, Pflueger & Viceira (2020): the sign of the bond–stock comovement follows the inflation–output relation and the monetary-policy regime
- Molenaar, Sénéchal, Swinkels & Wang (2024): evidence across countries and a long history
- The SPY–TLT correlation is positive on 81% of days since January 2022
- The Wied–Krämer–Dehling test dates the break on 3 August 2020
- Hypothesis with monthly realised correlations ρ_m: $z_m = \operatorname{artanh} \rho_m = \gamma_0 + \gamma_1 \pi_{m-2} + u_m$, π the annual US CPI (consumer price index) inflation known when month m starts, u_m the error
- H_0: $\gamma_1 = 0$ against H_1: $\gamma_1 > 0$, with a HAC t-statistic
- Falsified if $\gamma_1 \approx 0$ in periods where inflation varied but the regime did not
- The loop: an AI assistant drafts the literature map, competing hypotheses, the code, a robustness plan and a referee critique (prompts in the Appendix)
  - The researcher resolves every DOI, replicates the chapter's numbers first and fixes the plan before the first run

Mini-Case: Does Inflation Drive the Stock–Bond Correlation? (1/2)



MFM_ch6_ai_inflation

Mini-Case: Does Inflation Drive the Stock–Bond Correlation? (2/2)

- Monthly SPY–TLT correlation from daily returns against the inflation known at the start of the month (π_{m-2}: CPI for month m is published in month $m+1$)
- Current CPI vintage, so a retrospective test, not a real-time one
- Full sample, 287 months: $\hat{\gamma}_1 = 0.055$ per percentage point, HAC se 0.019, $t = 2.96$, $R^2 = 0.047$
- Mean ρ -0.12 when inflation exceeds 3% (89 months), -0.30 otherwise
- 2002–2021 only (232 months, 53 with inflation above 3%): $\hat{\gamma}_1 = 0.002$, $t = 0.06$: the full-sample slope rests on the post-2022 episode
- Associational, and one regime change cannot separate the inflation level from the policy regime (Campbell, Pflueger & Viceira (2020)): the claim is not yet established

MFM_ch6_ai_inflation

What the Researcher Checks, and a Project Seed

Checks before any claim

- Look-ahead bias: release calendar and data vintages (the lag handles publication delay, not revisions)
- Calendars: align prices on common trading days before computing returns
- Persistent regressor, few regimes: subsamples, long HAC lags, a placebo with simulated persistent series
- Volatility bias: control for realised volatility
- Multiple testing: fix measures and windows in advance
- Invented references: an unresolvable DOI is dropped and logged in `AI_ERRORS.md`

Your turn: a team-project seed

- Does a DCC whose target depends on inflation and the policy rate forecast the SPY–TLT correlation better out of sample than the plain DCC?
- Replicate first: $a + b = 0.986$, $\bar{Q}_{\mathbf{12}} = -0.25$ and the slope $\hat{\gamma}_{\mathbf{1}} = 0.055$
- Deliverables: a one-page pre-registration, code on the course market data plus CPI, a Diebold–Mariano comparison and a referee-style self-critique
- The AI-use statement, in `AI_USE.md`

Decision Checklist: Question, Estimator, Uncertainty, Failure

Question	Estimator	Uncertainty	Fails when
Did a correlation change?	Pearson on aligned prices	HAC delta method, block bootstrap	Fisher under clustering; daily data across time zones
Covariance for risk today?	EWMA, DCC	bootstrap of both steps	the target $\bar{Q}$ moves with the regime
When did it break?	Wied–Krämer–Dehling on GARCH residuals	Brownian-bridge limit	estimated GARCH residuals (approximate p-value); unmodelled variances
Contagion?	Forbes–Rigobon ρ^*	stationary bootstrap, δ estimated	a common shock or feedback
Joint extremes?	copula on GARCH pseudo-observations	rank sandwich, parametric bootstrap	Gaussian copula ($\lambda = 0$), static copula when dependence moves

- Next: Chapter 7 turns DCC covariances and simulated copulas into portfolio VaR and ES

Check Your Understanding

- Why is the Fisher interval for the SPY–TLT correlation too narrow on daily data, and which two features does the HAC interval take into account?
- Why is the same-day correlation of SPY and Euro Stoxx 50 biased towards zero, and how do 2-day or weekly returns reduce the bias?
- Which source of uncertainty do step-2 DCC standard errors ignore, and how does the bootstrap of both steps restore it?
- A crisis correlation is higher than the calm one: what must you compute before calling the rise contagion, and which assumption can make the adjustment excessive?
- Five copulas share Kendall's $\tau = 0.5$: why can their probabilities of joint crashes still differ?

Further Study — Courses, Code and Data

Advanced courses on quantinar.com

- Measuring Statistical Risk
- Multivariate Statistical Analysis
- Minimum Spanning Tree
- FRM: Financial Risk Meter

Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_06
- Textbooks: Franke, Härdle & Hafner (2019); Härdle & Simar (2019); Nelsen (2006)

Back

Inference on a Correlation, the Delta Method

- $\hat{\rho} = g(\hat{m})$, with $\hat{m}$ the sample mean of $m_t = (x_t,\, y_t,\, x_t^2,\, y_t^2,\, x_t y_t)$ and g the function that turns these five moments into the correlation
- ∇g: its gradient
- Delta method: $\sqrt{T}(\hat{\rho} - \rho) \to N\big(0,\ \nabla g^\top \Omega \nabla g\big)$, Ω the long-run covariance of m_t
- Requires stationarity, finite fourth moments of (x_t, y_t) and weak dependence (mixing)
  - $\hat{\Omega}$: HAC estimator with Bartlett kernel and L lags (Newey & West (1987))
  - Equivalent to the influence-function form $\hat{\rho} - \rho \approx \frac{1}{T}\sum_t \psi_t$ of the main text
- Fat tails typically raise $\mathrm{Var}(\psi_t)$ and GARCH can make ψ_t autocorrelated
- With infinite fourth moments the Normal limit fails
- Interval on the Fisher scale: $\tanh\big(\operatorname{artanh}\hat{\rho} \pm 1.96\,\widehat{\mathrm{se}}/(1-\hat{\rho}^2)\big)$
- Last window of the 5-day series (17 September 2026, one day before the last daily value, 0.24): $\hat{\rho} = 0.245$, Fisher [0.13, 0.36], HAC [0.07, 0.41]
- Largest width ratio 2.02

Back

The VEC and BEKK Models

- **VEC** (Bollerslev, Engle & Wooldridge (1988)): every element of H_t depends on all squares and cross-products
  - $\text{vech}(H_t) = c + A\,\text{vech}(r_{t-1}r_{t-1}^\top) + B\,\text{vech}(H_{t-1})$
  - vech: stacks the lower triangle of a symmetric matrix into a vector
  - c: constant vector
  - A, B: square parameter matrices
  - Positive definiteness is not guaranteed
  - Parameters: $N(N+1)/2 \cdot [1 + N(N+1)]$
- **BEKK** (Engle & Kroner (1995)): positive definite by construction if $CC^\top$ is positive definite
  - $H_t = CC^\top + A^\top r_{t-1}r_{t-1}^\top A + B^\top H_{t-1}B$
  - C: lower-triangular $N \times N$
  - A, B: full $N \times N$ matrices
  - Parameters: $N(N+1)/2 + 2N^2$
  - Hard to interpret, and the likelihood is flat in high dimension
- DCC keeps univariate GARCH for each variance and adds two correlation parameters: $3N + 2$ in total

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Two-Step Estimators, the Correct Covariance

- Stack the estimating equations: $\sum_t s_{1t}(\theta_1) = 0$ (step 1, GARCH) and $\sum_t s_{2t}(\theta_1, \theta_2) = 0$ (step 2, correlation)
- θ_1: GARCH parameters, $\theta_2 = (a, b)$, s_{jt}: score of step j on day t
- $A_{jk} = \mathbb{E}\left[\partial s_{jt} / \partial \theta_k^\top\right]$: A_{21} measures how the correlation score reacts to the GARCH parameters
- Newey & McFadden (1994), Section 6: $\sqrt{T}(\hat{\theta}_2 - \theta_2) \to N(0, V_2)$ with
  - $V_2 = A_{22}^{-1}\, \mathbb{E}\left[(s_{2t} - A_{21} A_{11}^{-1} s_{1t})(s_{2t} - A_{21} A_{11}^{-1} s_{1t})^\top\right] A_{22}^{-1\top}$
- Step-2 standard errors keep only s_{2t}: they drop $A_{21} A_{11}^{-1} s_{1t}$, the GARCH estimation error carried into step 2
- With correct conditional moments the scores are martingale differences, so the middle term is their variance
- Otherwise use its long-run (HAC) version
  - With targeting, $\bar{Q}$ is estimated too: its moment equations belong to step 1 (θ_1 includes vech $\bar{Q}$)
- Engle & Sheppard (2001) derive V_2 for DCC
- A simple alternative bootstraps the *whole* two-step procedure (next slide)

Back

Bootstrap of Both DCC Steps

- Parametric residual bootstrap, $B = 199$: decorrelate the residuals with the Cholesky factor of R_t, resample them and simulate DCC innovations with $(\hat{a}, \hat{b}, \bar{Q})$
- Step 2 only: re-estimate (a, b) on the simulated innovations
- Both steps: rebuild returns with the fitted GARCH, re-estimate GARCH, $\bar{Q}$ and (a, b) on the same innovations
- SPY–TLT: the step-2 Hessian se$(\hat{b})$ is 8% above the bootstrap se of both steps (0.00835 against 0.00772, unrounded)
- The CCC-null bootstrap of the LR test uses the same machinery with $a = b = 0$: constant-correlation innovations, simulated GARCH returns, both steps re-estimated

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Robert Engle and the DCC Model



Robert F. Engle at the Western Economic Association International conference, Santiago, 2017

Photo: Econterms (2017); CC BY-SA 4.0; Wikimedia Commons

- Nobel Memorial Prize in Economic Sciences 2003 for ARCH models (Chapter 5)
- Engle (2002): "Dynamic Conditional Correlation"
  - Keeps the flexibility of univariate GARCH for each variance
  - Adds a simple, estimable law of motion for correlations
- Refinement: the corrected DCC of Aielli (2013), which makes $\bar{Q}$ a consistent target

Back

Contagion Tests beyond Forbes–Rigobon

- Corsetti, Pericoli & Sbracia (2005): with a common factor the Forbes–Rigobon adjustment over-corrects, biasing the test towards "no contagion"
- They give a factor-model version
- Rigobon (2003): identification through heteroskedasticity
- The variance shift between regimes identifies the structural coefficient β, an instrumental-variables idea
- With δ estimated by the bootstrap of the main text the 2008 verdict stands
- The common-factor critique remains (Seminar A1, B3)

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Copula Densities and Constructions

- Gaussian density: $c = \frac{1}{\sqrt{1-\rho^2}} \exp\left(- \frac{\rho^2(x^2+y^2)-2\rho xy}{2(1-\rho^2)} \right)$, $x = \Phi^{-1}(u)$, $y = \Phi^{-1}(v)$
- Symmetric, no tail dependence for $|\rho| < 1$
- Student t: small ν means strong joint extremes, equal in both tails
- The t copula is the building block of risk systems with a correlation matrix
- Archimedean generators φ in $C(u,v) = \varphi^{-1}(\varphi(u) + \varphi(v))$
  - Clayton $\varphi(t) = (t^{-\theta} - 1)/\theta$
  - Gumbel $\varphi(t) = (-\ln t)^{\theta}$
  - Frank $\varphi(t) = -\ln \frac{e^{-\theta t}-1}{e^{-\theta}-1}$
- Asymmetric tails: Clayton for joint crashes of returns, Gumbel for joint extremes of losses
- The 180° rotation $\bar{C}(u,v) = u + v - 1 + C(1-u, 1-v)$ moves the tail to the opposite corner

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Copula Estimates for JPM and BAC, Full Table

Copula	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	ℓ	ΔAIC	λ_L	λ_U	fit p
Gaussian	0.833	–	2 486.0	309.3	0.000	0.000	0.005
Student t	0.842	4.24	2 641.7	0.0	0.531	0.531	0.010
Clayton	2.384	–	2 116.7	1 048.0	0.748	0.000	0.005
Gumbel	2.587	–	2 412.2	456.9	0.000	0.693	0.005
Frank	9.152	–	2 390.8	499.8	0.000	0.000	0.005

- CML on pseudo-observations of GARCH residuals
- ΔAIC relative to the best model
- Goodness-of-fit p from the Rosenblatt test with parametric bootstrap

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Rank Estimators and Their Precision

- $\hat{\tau}$ is a U-statistic
- Its Hoeffding projection gives $\sqrt{n}(\hat{\tau} - \tau) \to N\big(0,\ 16\mathrm{Var}[2C(U,V) - U - V]\big)$
- Plug-in: $\widehat{\mathrm{se}}(\hat{\tau}) = 4\,\mathrm{sd}\big(2\hat{C}_n(\hat{U}_i, \hat{V}_i) - \hat{U}_i - \hat{V}_i\big)/\sqrt{n}$
- Inversion estimators $\hat{\theta} = g(\hat{\tau})$ by the delta method (Genest & Rivest (1993))
- The moment estimator is almost as precise as CML for ρ and is a robust starting value for the likelihood
- CML sandwich (Genest, Ghoudi & Rivest (1995)): $\sqrt{n}(\hat{\theta} - \theta) \to N(0,\ A^{-1}\Sigma A^{-1})$, A the Fisher information of the copula, $\Sigma = \mathrm{Var}\big[\dot{\ell}_\theta(U,V) + W_1(U) + W_2(V)\big]$, $\dot{\ell}_\theta$ the score of $\ln c$
  - $W_1(u) = \iint 1\{u \le u'\}\,\partial_u \dot{\ell}_\theta(u', v')\,dC(u', v')$: the price of ranking
  - W_2 likewise in v
  - Gaussian copula: variance $(1-\rho^2)^2$ for CML, against $(1-\rho^2)^2/(1+\rho^2)$ with known margins
- S&P 500–Euro Stoxx 50 weekly, Gaussian: se 0.0086 (inverse Hessian) versus 0.0138 (sandwich)
- JPM–BAC, t: ρ 0.0046 versus 0.0053, ν 0.359 versus 0.364

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Chen and Fan (2006), GARCH Filtering and the Limit

- Model: $r_{it} = \mu_i + \sigma_{it}\varepsilon_{it}$ with GARCH σ_{it}
- Innovations $(\varepsilon_{1t}, \varepsilon_{2t})$ i.i.d. with copula C_θ
- Chen & Fan (2006): CML on the ranks of the *estimated* residuals $\hat{\varepsilon}_{it}$ has the same limit distribution as CML on the ranks of the true ε_{it}
- Monte Carlo: 500 samples, $T = 2\,000$, Gaussian copula $\rho = 0.8$, Student t(5) innovations, GARCH as fitted to JPM and BAC
  - Standard deviation of $\hat{\rho}$: 0.0081 (true innovations) and 0.0082 (GARCH residuals)
  - Mean sandwich se 0.0081, mean inverse-Hessian se 0.0063

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Nonparametric Tail Dependence, Conditions

- Schmidt & Stadtmüller (2006): the estimator $\hat{\lambda}_L$ of the main text is consistent and asymptotically Normal if $k \to \infty$, $k/n \to 0$, the tail copula exists and a second-order bias condition holds ($\sqrt{k}\, A(n/k) \to 0$, $A(\cdot)$ the rate at which $C(q,q)/q$ approaches λ_L)
- The i.i.d. bootstrap of residual pairs assumes a static innovation copula
- The stationary block bootstrap allows serial dependence in the tail indicators $1\{\text{lower corner}\} - 1\{\text{upper corner}\}$
- Mean block length by the automatic rule of Politis & White (2004), applied to the tail indicator
- $B = 999$ draws, ranks recomputed in each draw

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Dependence in High Dimension and over Time

- DCC for large N: the full likelihood is slow and biased
  - Composite likelihood over pairs of assets, Pakel, Shephard, Sheppard & Engle (2021)
  - DCC-NL, nonlinear shrinkage of the target $\bar{Q}$ (the shrinkage of Chapter 4), Engle, Ledoit & Wolf (2019)
- Vine (pair-copula) constructions, Aas, Czado, Frigessi & Bakken (2009)
  - $d(d-1)/2$ bivariate copulas, some conditional, in nested trees
  - Each pair has its own family and tails
- Hierarchical Archimedean copulas, Okhrin, Okhrin & Schmid (2013)
  - Groups nested by the strength of dependence: banks within a country, countries within a region
- Dynamic copulas
  - Score-driven parameters (Creal, Koopman & Lucas (2013)) extend to many assets through factor copulas
  - Oh & Patton (2018) measure time-varying systemic risk in credit default swap spreads
  - Conditional copulas: Patton (2006)

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From Correlations to Systemic Risk

- Correlation networks are the first step towards the systemic-risk measures of Chapter 18
  - Connectedness from Granger causality: Billio, Getmansky, Lo & Pelizzon (2012)
  - Spillover indices from variance decompositions: Diebold & Yilmaz (2009)
  - Tail-event networks: the Financial Risk Meter applied to EU energy firms, Toma, Dima, Grecu, Pele & Lessmann (2026)
- Energy prices and EU stock markets: dependence across regions and industrial groups, Grecu, Cramer, Pele & Lessmann (2025)
- Correlation networks inherit all the problems of correlations: non-linearity, time variation, tails

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COVOL: Data and Estimation

- **Model** (Eq. 8–11 of Engle & Campos-Martins (2023)): for $i \neq j$, $\Psi_{ij,t} = s_i s_j v_t$, $v_t = \mathrm{Var}_{t-\mathbf{1}}(x_t)$, so Ψ has one factor and its first principal component gives starting values (Eq. 12–13)
  - Equal loadings $s_i = 1$: the cross-sectional mean $\hat{x}_t$ of the main text is the maximum-likelihood estimate (Eq. 17)
- **Estimation** (Section 5.3): maximise $-\frac{\mathbf{1}}{\mathbf{2}} \sum_{i,t} \left[\ln g_{it} + e_{it}^{\mathbf{2}} / g_{it} \right]$, $g_{it} = s_i x_t + 1 - s_i$, alternating x_t and s_i until convergence
  - After each step: $s^\top s = 1$, mean of x_t equal to 1 (Remark 2)
  - Test ξ (Eq. 27): standardised mean of $(e_{it}^{\mathbf{2}} - 1)(e_{jt}^{\mathbf{2}} - 1)$ over the $N(N-1)/2$ pairs, $\mathcal{N}(0,1)$ under $H_{\mathbf{0}}$
- **Data** (Section 7.1, Table B.1): daily USD log returns of 47 country ETFs listed on the NYSE, 2 June 2000 – 1 March 2021
- Same closing time, so no asynchrony
- Unbalanced panel
  - Mean: AR(1) plus two factors, the ACWI and PC1 of the ETF returns
  - GARCH(1,1) variances
  - COVOL fitted to the 47 residual series and the two factors
  - Also: 23 US asset-class series (Section 9) and Monte Carlo studies of the estimator and of the test (Sections 5.4, 6.1)

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COVOL: Key Results

Day	$\hat{x}_t$	$\bar{r}_t$ (%)	Event (label of the paper)
9 March 2020	51.416	−9.650	COVID-19: Italy locked down; oil price war
24 June 2016	49.190	−7.270	Brexit
5 August 2019	39.457	−2.574	Trade war: China devalues its currency
28 April 2020	38.883	2.334	COVID-19: tech share sell-off
17 September 2001	30.077	−8.630	NYSE reopens after 9/11

$\bar{r}_t$: mean return across the country ETFs; top five of the twenty largest values in Table 6 of Engle & Campos-Martins (2023)

- Loadings (Table 7, normalised by $s^\top s = 1$, so relative sizes, not variance shares): PC1 0.261 and ACWI 0.250 lead
- France 0.217, Spain 0.200 against New Zealand 0.065, Poland 0.080: factors load more than idiosyncrasies, so correlations rise with COVOL (Eq. 41)
- ACWI squared residual on COVOL_m^2 (monthly mean of $\hat{x}_t$ minus 1): 0.326 (se 0.025), $R^2 = 0.404$ (Table 8)
- Mean pairwise ETF correlation +0.053 per unit (se 0.009, Table 10, column 4)
- Change in VIX (Cboe index of the implied volatility of the S&P 500) 4.034 (se 0.249), $R^2 = 0.517$ (Table 12)

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COVOL: Replication Protocol and Limitations

- **Replicate** the asset-class version (Section 9) with the US-listed ETFs from the course market data, which all close at the same time: 11 sector ETFs, Russell 1000 growth and value, Russell 2000, developed and emerging markets outside the US, gold, long Treasuries, corporate bonds, crude oil
  - One factor, PC1
  - Unbalanced panel
  - $\hat{x}_t$ by Eq. 17, then the iteration of Section 5.3
  - ξ before and after dividing e_{it}^2 by $\hat{g}_{it}$ (Section 7.2)
  - Benchmark: Table 15 ranks 9 November 2016 (43.04) and 9 March 2020 (40.47) among the top four
  - The dollar and the Russell 2000 growth/value split are not in the course data
- **Limitations stated by the authors**
  - After dividing by $\hat{g}_{it}$ the squares are still correlated: possibly a second factor (Section 7.2)
  - Multifactor COVOL is open (Section 9)
  - With $s^\top s = 1$ the loadings shrink as N grows: $R_s^2 = 0.640$ for $N = 50$, $T = 1000$ against 0.924 for $N = 10$ (Table 2)
  - The test loses power with unequal loadings, so $T \geq 1000$ is advised
- After the sample of the paper: 3 April 2025 (16.2, -3.67%), after the US tariff announcement, and 10 November 2022 (15.4, 4.56%), a lower-than-expected US inflation print
  - Open question (Seminar C3): do elections, tariffs and inflation news share one factor?
  - Open question: does COVOL_m^2 explain the monthly DCC correlations beyond the change in VIX?

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The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**: *"List studies since 2015 on the drivers of the stock–bond correlation, with DOI, sample, measure of correlation, identification strategy and main estimate."*
  - Resolve every DOI on Crossref and check the estimate in the paper, not in the summary
- **Hypotheses**: *"Give competing explanations for the 2022 sign change (inflation level, monetary-policy regime, term-premium shock, a one-off tightening) and what each predicts for ρ before 2022 and out of sample."*
- **Data and code**: *"Write Python that aligns SPY and TLT adjusted prices from the course market data on common trading days, computes monthly realised correlations and merges US CPI inflation published before each month starts."*
  - Replicate first: -0.40 (2002–2021) and 0.11 (2022–2026)

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**: *"Write a pre-registered plan: the pre-2022 subsample, core CPI instead of headline, weekly instead of daily returns, DCC instead of realised correlation, control for realised volatility, HAC lag 12."*
  - Placebo: replace π by simulated random walks of the same persistence
  - A persistent regressor can produce large t-statistics by chance
- **Critique**: *"Act as a referee: is γ_1 identified by a single episode, and would the Forbes–Rigobon volatility bias explain it? Give the test that would expose each."*
- **Report**: every inflation measure, window and frequency tried
- Subsample estimates next to the full sample
- Any specification chosen after seeing results is exploratory and reported as such
- Code of the mini-case: Quantlets/Ch_06/ai_discovery_case.py
- Inflation: US consumer price index for all urban consumers, U.S. Bureau of Labor Statistics

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Acronyms used in this material (1/2)

- **ACWI**: All Country World Index (MSCI)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **BAC**: Bank of America Corporation (NYSE ticker)
- **BEKK**: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNP**: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- **CCC**: Constant Conditional Correlation (model)
- **CDO**: Collateralised Debt Obligation
- **CML**: Canonical Maximum Likelihood
- **COVID-19**: COronaVIrus Disease 2019
- **COVOL**: COmmon VOLatility (global COVOL: the size of the volatility shock common to all assets)
- **CPI**: Consumer Price Index
- **DBK**: Deutsche Bank AG (Xetra ticker)
- **DCC**: Dynamic Conditional Correlation
- **DCC-NL**: Dynamic Conditional Correlation with NonLinear shrinkage

Acronyms used in this material (2/2)

- **EPU**: Economic Policy Uncertainty (index)
- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund
- **EU**: European Union
- **EWMA**: Exponentially Weighted Moving Average
- **FRM**: Financial Risk Meter
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GAS**: Generalised Autoregressive Score (model)
- **GPR**: GeoPolitical Risk (index)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **IFM**: Inference Functions for Margins
- **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- **LR**: Likelihood Ratio (test)
- **MSCI**: Morgan Stanley Capital International (index provider)
- **NYSE**: New York Stock Exchange
- **PC1**: first Principal Component
- **US**: United States
- **VEC**: VECtorised multivariate GARCH model (vech of the covariance matrix)
- **VIX**: CBOE Volatility IndeX

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